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The Brownian Map

An Exploration of a Quotient of a Random Tree

Semester Thesis

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Abstract

This thesis delves into the rich mathematical terrain of the continuum random trees (CRT) and its intriguing connection to the Brownian map, with a particular focus on the homeomorphism of the Brownian map onto the 2-sphere. The study begins by providing a comprehensive overview of real trees and the continuum random tree, elucidating their construction and relation to finite discrete trees. As a next step, the construction of the Brownian map via Brownian motion indexed by the continuum random tree will be explored. Finally, the homeomorphism between the Brownian map and the 2-sphere is uncovered. Along the road, all the mathematical tools, starting from introduction of metric spaces and the Gromov-Hausdorff distance, over the rich theory of real trees and their stochastic counterparts, to the discussion of hyperbolic geodesic laminations and their connection to quotients of real trees, will be introduced.

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Chapter 1

Metric Spaces and the Gromov-Hausdorff Distance

1.1 Metric spaces

In this section I introduce the definition and some properties of metric spaces, which will be useful later on when defining $\mathbb{R}$ -trees and discussing the *Brownian map*. This section closely follows the book [1].

Definition 1.1 A *metric space* is a set X together with a map $d : X \times X \rightarrow \mathbb{R}$ called *metric* such that for all $x, y, z \in X$ the following holds.

- (i) **Positiveness:** $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
- (ii) **Symmetry:** $d(x, y) = d(y, x)$.
- (iii) **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

I will denote for any $\delta > 0$ the open ball of radius δ about $x \in X$ by

$$B_\delta(x) = \{y \in X \mid d(x, y) < \delta\}.$$

Moreover, if (X, d) only satisfies (ii), (iii) and

- (i') $d(x, y) \geq 0$,

we call X a *pseudo-metric space* and d a *pseudo-metric*.

Remark 1.2 It follows quite easily from the properties (i) – (iii) of a metric space (X, d) that for all $x, y, z \in X$ the so-called **reverse triangle inequality** given by

$$|d(x, y) - d(y, z)| \leq d(x, z)$$

holds.

Now I will give a short proof on why in a metric space (X, d) the mapping d is a continuous map $X \times X \rightarrow \mathbb{R}$.

Proposition 1.3 *Let (X, d) be a metric space. The metric $d : X \times X \rightarrow \mathbb{R}$ is continuous with respect to the product topology on $X \times X$ and the standard topology on $\mathbb{R}$.*

Proof Let $\varepsilon > 0$ and $(z_0, w_0) \in X \times X$. For $\delta = \frac{\varepsilon}{2}$ set $U = B_\delta(z_0) \times B_\delta(w_0)$. For all $(z, w) \in U$ it holds by the triangle and reverse triangle inequality that

$$\begin{aligned} |d(z, w) - d(z_0, w_0)| &\leq |d(z, w) - d(w, z_0)| + |d(w, z_0) - d(z_0, w_0)| \\ &\leq d(z, z_0) + d(w, w_0) \\ &< 2\delta = \varepsilon. \end{aligned}$$

This concludes the proof. $\square$

Of particular interest in this thesis are metric-preserving morphisms between metric spaces, the so-called *isometries*.

Definition 1.4 *Let (X, d_X) and (Y, d_Y) be metric spaces. A continuous bijection $f : X \rightarrow Y$ is called an isometry if for all $x, y \in X$*

$$d_Y(f(x), f(y)) = d_X(x, y).$$

*If there exists an isometry $X \rightarrow Y$, we call X and Y **isometric**.*

I now give the construction of a metric space starting from a pseudometric space. This construction will be fundamental in the thesis, when constructing *real trees*.

Construction 1.5 *Let (X, d) be a pseudometric space and define an equivalence relation $\sim$ on X as follows:*

$$x \sim y \iff d(x, y) = 0.$$

Define $\bar{X} := X/\sim$, the set of equivalence classes, and the map $\bar{d} : \bar{X} \times \bar{X} \rightarrow \mathbb{R}$ by $\bar{d}([x], [y]) = d(x, y)$. This is clearly well-defined as if $([x], [y]) = ([x'], [y'])$, then $d(x, x') = d(y, y') = 0$. Hence, by the triangle inequality of d , we find

$$\begin{aligned} \bar{d}([x], [y]) &= d(x, y) \\ &\leq d(x, x') + d(x', y) \\ &\leq d(x', y') + d(y', y) \\ &= \bar{d}([x'], [y']). \end{aligned}$$

Similarly, we find that $\bar{d}([x'], [y']) \leq \bar{d}([x], [y])$. Thus, $\bar{d}([x'], [y']) = \bar{d}([x], [y])$. The positiveness, symmetry, and triangle inequality of $\bar{d}$ follow directly from the properties of the pseudometric d . For the definiteness, we note that for all $[x], [y] \in \bar{X}$

$$\begin{aligned} \bar{d}([x], [y]) = 0 &\iff d(x, y) = 0 \\ &\iff x \sim y \\ &\iff [x] = [y]. \end{aligned}$$

This shows that $(\overline{X}, \overline{d})$ is a metric space.

1.1.1 Hausdorff distance and Gromov-Hausdorff distance

An important example of a metric used in this thesis is the so-called **Hausdorff distance** on non-empty compact subsets of a metric space (X, d) . For this example, we first need the following corollary to Proposition 1.3.

Corollary 1.6 *Let (X, d) be a metric space and $A \subset X$ a non-empty compact subset. The map*

$$x \in X \mapsto d(x, A) := \inf\{d(x, a) \mid a \in A\}$$

is continuous.

Proof By Proposition 1.3 the metric d is continuous; hence, for any $x \in X$ the map $f_x : X \rightarrow \mathbb{R}$ given by $f_x(y) = d(x, y)$ is also continuous. By compactness of A , the continuous map f_x attains a minimum on A , i.e., there exists $a \in A$ s.t. $d(x, A) = f_x(a) = d(x, a)$. Hence, for every $x, y \in X$ we find $a, b \in A$ such that $d(x, A) = d(x, a)$ and $d(y, A) = d(y, b)$, and so

$$\begin{aligned} d(y, A) &= d(y, b) \\ &\leq d(x, y) + d(x, b). \end{aligned}$$

By taking the infimum on over all $b \in A$ we find $d(y, A) - d(x, A) \leq d(x, y)$. Similarly, we have $d(x, A) - d(y, A) \leq d(x, y)$. Thus, we have that

$$|d(x, A) - d(y, A)| \leq d(x, y).$$

So $d(\cdot, A)$ defines a 1-Lipschitz continuous map. In particular, it is continuous. $\square$

Now, I will present the example of the **Hausdorff distance**.

Example 1.7 *Let (X, d) be a metric space. Define for any two non-empty compact subsets $A, B \subset X$:*

$$d_H(A, B) := \max\{\max\{d(b, A) \mid b \in B\}, \max\{d(a, B) \mid a \in A\}\}.$$

*We call $d_H(A, B)$ the **Hausdorff distance** between A and B .*

In the next lemma, I prove that the Hausdorff distance d_H actually defines a metric on the set of non-empty compact subsets of a metric space.

Lemma 1.8 *Let (X, d) be a metric space. The Hausdorff distance d_H as previously defined in 1.7 defines a metric on the set of non-empty compact subsets of X .*

Proof We check for all non-empty compact subsets $A, B, C \subset X$:

- (i) **Positiveness:** By definition $d(\cdot, A) \geq 0$ for any nonempty compact subset $A \subset X$, so that $d_H(A, B) \geq 0$. Now if $d_H(A, B) = 0$, then we find $d(a, B) = d(b, A) = 0$ for all $a \in A$ and $b \in B$. Because $d(x, A) = 0$ if and only if $x \in A$ by compactness of A , it follows that $A = B$. The converse is trivial.
- (ii) **Symmetry:** By the symmetry of $\max\{\cdot, \cdot\}$ the symmetry of d_H follows.
- (iii) **Triangle inequality:** For all $a \in A, b \in B, c \in C$ we have that

$$\begin{aligned} d(a, C) &\leq d(a, c) \\ &\leq d(a, b) + d(b, c). \end{aligned}$$

By taking the infimum over all $b \in B$ we find that $d(a, C) \leq d(a, B) + d(c, B)$. Similarly, we find $d(c, A) \leq d(a, B) + d(c, B)$. Hence, by definition of d_H we get

$$d_H(A, C) \leq d_H(A, B) + d_H(B, C).$$

This shows that the set of non-empty compact subsets of X together with the Hausdorff distance d_H define a metric space. $\square$

From this example, we can extract another example of a metric space, the set of compact metric spaces together with the so-called **Gromov-Hausdorff distance**. The following example is derived from definition given in [2].

Example 1.9 For two compact metric spaces (X, d_X) and (Y, d_Y) define

$$\tilde{d}_{GH}(X, Y) := \inf \left\{ d_H(\varphi(X), \psi(Y)) \left| \begin{array}{l} \varphi : X \rightarrow W, \psi : Y \rightarrow W \\ \text{are isometric embeddings} \\ \text{to metric space } (W, d) \end{array} \right. \right\}.$$

The positiveness, symmetry, and triangle inequality of $\tilde{d}_{GH}$ follow from the properties of the Hausdorff distance. Define now $\mathbb{K}$ to be the set of (isometry) equivalence classes of compact metric spaces under the equivalence relation on compact metric spaces defined by

$$(X, d_X) \approx (Y, d_Y) \iff X \text{ and } Y \text{ are isometric.}$$

Now, define for all equivalence classes $[X]_{\approx}, [Y]_{\approx} \in \mathbb{K}$ the **Gromov-Hausdorff distance** by

$$d_{GH}([X]_{\approx}, [Y]_{\approx}) := \tilde{d}_{GH}(X, Y).$$

This map is well-defined by the same logic as in the Construction 1.5. Positiveness, symmetry, and triangle inequality also follow from $\tilde{d}_{GH}$. It remains to shown to be definite.

Remark 1.10 By abuse of notation, I denote an isometry class $[X]_{\approx} \in \mathbb{K}$ also by X for all non-empty compact metric spaces X .

To show that this is actually a metric requires a reformulation. I will follow Chapter 7 of [1].

Definition 1.11 Let (X, d_X) and (Y, d_Y) be metric spaces and $\mathcal{R} \subset X \times Y$ be a bisurjective relation¹ Define the **distortion** of $\mathcal{R}$ by

$$\text{dis}\mathcal{R} := \sup\{|d_X(x, x') - d_Y(y, y')| \mid (x, y), (x', y') \in \mathcal{R}\}.$$

The next proposition provides a link between the Gromov-Hausdorff distance and distortions.

Proposition 1.12 For any two compact metric spaces (X, d_X) and (Y, d_Y) it holds that

$$\tilde{d}_{GH}(X, Y) = \frac{1}{2} \inf_{\mathcal{R}} (\text{dis}\mathcal{R}),$$

where the infimum is taken over the set of bisurjective relations of X and Y

Proof Set $D = \tilde{d}_{GH}(X, Y)$. Let $\varepsilon > 0$ then there exists a metric space (W, d) and isometries $f : X \rightarrow W$ and $g : Y \rightarrow W$ such that

$$d_H(f(X), g(Y)) < D + \varepsilon.$$

Now define

$$\mathcal{R} := \{(x, y) \in X \times Y \mid d(f(x), g(y)) < D + \varepsilon\}.$$

Since $d_H(f(X), g(Y)) < D + \varepsilon$ it follows that $\mathcal{R}$ is bisurjective relation. Note now that we have for all $(x, y), (x', y') \in \mathcal{R}$ that

$$\begin{aligned} |d_X(x, x') - d_Y(y, y')| &= |d(f(x), f(x')) - d(g(y), g(y'))| \\ &\leq |d(f(x), f(x')) - d(f(x'), g(y))| \\ &\quad + |d(f(x'), g(y)) - d(g(y), g(y'))| \\ &\leq d(f(x), g(y)) + d(f(x'), g(y')) \\ &< 2(D + \varepsilon). \end{aligned}$$

By taking supremum and letting $\varepsilon \rightarrow 0$ we find $\frac{1}{2} \text{dis}\mathcal{R} \leq D$. In particular $\frac{1}{2} \inf_{\mathcal{R}} (\text{dis}\mathcal{R}) \leq D$. For the converse inequality, consider any bisurjective relation $\mathcal{R}$ on $X \times Y$ with $\delta = \text{dis}\mathcal{R}$. Define now on the disjoint union $X \sqcup Y$ the following map

$$d(x, y) = \inf \left\{ d_X(x, x') + d_Y(y, y') + \frac{\delta}{2} \mid (x', y') \in \mathcal{R} \right\},$$

¹For any $x \in X$ there exists $y \in Y$ such that $(x, y) \in \mathcal{R}$, and similarly for any $y \in Y$ there exists $x \in X$ such that $(x, y) \in \mathcal{R}$.

where $x \in X$ and $y \in Y$ and setting $d|_X = d_X$ and $d|_Y = d_Y$. Positiveness, symmetry, and triangle inequality follow directly from the metrics d_X, d_Y . Up to taking the quotient by the equivalence relation $x \sim y \iff d(x, y) = 0$, we can consider the metric space $(X \sqcup Y, d)$ and the isometric inclusions $X, Y \hookrightarrow X \sqcup Y$. Note that for $(x, y) \in \mathcal{R}$ we have $d(x, y) = \frac{\delta}{2}$. Then clearly for all $x \in X$ and $y \in Y$ it follows that $d(x, Y) \leq d(x, y) \leq \frac{\delta}{2}$ and $d(y, X) \leq \frac{\delta}{2}$. Therefore, $d_H(X, Y) \leq \frac{\delta}{2}$, where we identify X, Y by their isometric images in $X \sqcup Y$. Since $\mathcal{R}$ was arbitrary it follows that $D \leq \frac{1}{2} \inf_{\mathcal{R}}(\text{dis}\mathcal{R})$. $\square$

Definition 1.13 Let (X, d_X) and (Y, d_Y) be compact metric spaces and $\varepsilon > 0$. A map $f : X \rightarrow Y$ is a ε -**isometry** if

- (i) $\text{dis}f := \sup\{|d_Y(f(x), f(y)) - d_X(x, y)| \mid x, y \in X\} < \varepsilon$ and
- (ii) $d_Y(y, f(X)) \leq \varepsilon$ for all $y \in Y$.

Remark 1.14 Note that for a surjective map $f : X \rightarrow Y$ we can define a bisurjective relation $\mathcal{R}_f := \{(x, f(x)) \mid x \in X\}$. Then clearly $\text{dis}\mathcal{R} = \text{dis}f$.

Corollary 1.15 Let (X, d_X) and (Y, d_Y) be compact metric spaces and $\varepsilon > 0$. Then:

- (i) If $\tilde{d}_{GH}(X, Y) < \varepsilon$, then there exists a 2ε -isometry from X to Y
- (ii) If there exists a ε -isometry from X to Y , then $\tilde{d}_{GH}(X, Y) < 2\varepsilon$.

Proof (i) Since $\tilde{d}_{GH}(X, Y) = \frac{1}{2} \inf_{\mathcal{R}}(\text{dis}\mathcal{R})$ we can choose a bisurjective relation $\mathcal{R}$ between X and Y such that $\text{dis}\mathcal{R} < 2\varepsilon$. By bisurjectivity for any $x \in X$ choose $f(x) \in Y$ such that $(x, f(x)) \in \mathcal{R}$. Therefore, $f : X \rightarrow Y$ is a map with $\text{dis}f < 2\varepsilon$ by Remark 1.14 and since $\mathcal{R}_f \subset \mathcal{R}$. For any $y \in Y$ pick now $x \in X$ such that $(x, y) \in \mathcal{R}$. Note now that because $(x, y), (x, f(x)) \in \mathcal{R}$ and $d_X(x, x) = 0$, we have

$$d_Y(y, f(x)) \leq \text{dis}\mathcal{R} < 2\varepsilon.$$

Hence, we find that $d(y, f(X)) < 2\varepsilon$, i.e. f is a 2ε -isometry.

- (ii) Suppose $f : X \rightarrow Y$ is a ε -isometry and define

$$\mathcal{R} := \{(x, y) \in X \times Y \mid d_Y(y, f(x)) \leq \varepsilon\}.$$

Because $d(y, f(X)) \leq \varepsilon$, it follows that $\mathcal{R}$ is a bisurjective relation. For any $(x, y), (x', y') \in \mathcal{R}$ we have that

$$\begin{aligned} |d_Y(y, y') - d_X(x, x')| &\leq |d_Y(f(x), f(x')) - d_X(x, x')| \\ &\quad + d(y, f(x)) + d(y', f(x')) \\ &\leq \text{dis}f + \varepsilon + \varepsilon \\ &\leq 3\varepsilon. \end{aligned} \quad \square$$

Hence, we find that $\text{dis}\mathcal{R} \leq 3\varepsilon$. Hence by Proposition 1.12 it follows that

$$\tilde{d}_{GH}(X, Y) \leq \frac{3}{2}\varepsilon < 2\varepsilon.$$

Proposition 1.16 *The Gromov-Hausdorff distance d_{GH} on the isometry classes $\mathbb{K}$ of compact metric spaces defines a metric.*

Proof By Example 1.9 it suffices to prove that for any compact metric spaces (X, d_X) and (Y, d_Y) it holds that $\tilde{d}_{GH}(X, Y) = 0$ if and only if X and Y are isometric. If X and Y are isometric then there exists an isometry $f : X \rightarrow Y$ such that $f(X) = Y$. Then, it follows that by definition of the Gromov-Hausdorff distance that

$$\tilde{d}_{GH}(X, Y) \leq d_H(f(X), Y) = d_H(Y, Y) = 0.$$

Hence, $\tilde{d}_{GH}(X, Y) = 0$. Conversely, if $\tilde{d}_{GH}(X, Y) = 0$, we have $\tilde{d}_{GH}(X, Y) < \frac{1}{2n}$ for all $n \in \mathbb{N}$. Therefore, by Corollary 1.15 for any $n \in \mathbb{N}$ we find a $\frac{1}{n}$ -isometry $f_n : X \rightarrow Y$. Therefore, $\text{dis}f_n \rightarrow 0$ as $n \rightarrow \infty$. By compactness, X admits a countable dense subset² $S = \{x_m\}_{m \in \mathbb{N}}$. Now, by compactness of Y , we can extract a convergent subsequence of $\{f_n(x_1)\}_{n \in \mathbb{N}}$, denote this subsequence by $\{f_{1,n}(x_1)\}_{n \in \mathbb{N}}$. Similarly, we can extract a convergent subsequence of $\{f_{1,n}(x_2)\}_{n \in \mathbb{N}}$, which we denote by $\{f_{2,n}(x_2)\}_{n \in \mathbb{N}}$. We then recursively find for each $k \in \mathbb{N}$ a convergent subsequence $\{f_{k,n}(x_k)\}_{n \in \mathbb{N}}$ of all sequences $\{f_{j,n}(x_j)\}_{n \in \mathbb{N}}$ for $j < k$. Set now $f(x_j) := \lim_{n \rightarrow \infty} f_{j,n}(x_j)$ for all $j \in \mathbb{N}$. Then again $\{f_{n,n}(x_j)\}_{n \in \mathbb{N}}$ is a convergent subsequence of all the sequences before such that $\lim_{n \rightarrow \infty} f_{n,n}(x_j) = f(x_j)$ for any $j \in \mathbb{N}$. Up to restricting to this convergent subsequence, we can assume that $\{f_n(x)\}_{n \in \mathbb{N}}$ converges in Y for any $x \in S$. Therefore, we can define a map $f : S \rightarrow Y$ by setting $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for any $x \in S$. Because we have for any $x, y \in S$

$$|d_Y(f_n(x), f_n(y)) - d_X(x, y)| \leq \text{dis}f_n \rightarrow 0$$

as $n \rightarrow \infty$, it follows from the continuity of d_Y that

$$d_Y(f(x), f(y)) = \lim_{n \rightarrow \infty} d_Y(f_n(x), f_n(y)) = d_X(x, y).$$

Hence, $f : S \rightarrow Y$ is isometric embedding. Because S is dense for any $x \in X$ we find a sequence $\{s_n\}_{n \in \mathbb{N}}$ in S such that $d_X(s_n, x) \rightarrow 0$ as $n \rightarrow \infty$. By the

²This is a direct consequence of compactness: Consider for each $n \in \mathbb{N}$ the open cover $\{B_{\frac{1}{n}}(x)\}_{x \in X}$. Because of the compactness of X , for each $n \in \mathbb{N}$ there is a finite subset $S_n \subset X$ such that $\{B_{\frac{1}{n}}(x)\}_{x \in S_n}$ is a finite subcover of X . Then, the countable union of finite sets $S = \bigcup_{n \in \mathbb{N}} S_n$ is countable. Now, for any $x \in X$ and $\varepsilon > 0$ we find $n \in \mathbb{N}$ and $y \in S_n$ such that $\frac{1}{n} < \varepsilon$ and $x \in B_{\frac{1}{n}}(y)$. As $d_X(x, y) < \frac{1}{n} < \varepsilon$ we have in particular $B_\varepsilon(x) \cap S \neq \emptyset$, as $y \in S \cap B_\varepsilon(x)$. This shows density.

continuity, we must have that $\{f(s_n)\}_{n \in \mathbb{N}}$ converges in Y . Define now $\tilde{f}(x) = \lim_{n \rightarrow \infty} f(s_n)$. This clearly defines a map in $\tilde{f} : X \rightarrow Y$. This map is well-defined because it does not depend on the choice of the sequence $\{s_n\}_{n \in \mathbb{N}}$ in S : Picking another sequence $\{t_n\}_{n \in \mathbb{N}}$ in S such that $d_X(t_n, x) \rightarrow 0$ as $n \rightarrow \infty$ gives us that as $n \rightarrow \infty$

$$\begin{aligned} d_Y(f(t_n), f(s_n)) &= d_X(t_n, s_n) \\ &\leq d_X(t_n, x) + d_X(s_n, x) \rightarrow 0. \end{aligned}$$

If now $x, y \in X$ we can choose sequences $\{s_n\}_{n \in \mathbb{N}}$ and $\{t_n\}_{n \in \mathbb{N}}$ in S such that $d_X(s_n, x), d_X(t_n, y) \rightarrow 0$ as $n \rightarrow \infty$ and obtain by continuity of a metric that

$$\begin{aligned} d_Y(\tilde{f}(x), \tilde{f}(y)) &= \lim_{n \rightarrow \infty} d_Y(f(s_n), f(t_n)) \\ &= \lim_{n \rightarrow \infty} d_X(s_n, t_n) \\ &= d_X(x, y). \end{aligned}$$

This shows that $\tilde{f} : X \rightarrow Y$ is an isometric embedding. Similarly, by exchanging the roles of X and Y in the above argument, we can construct an isometric embedding $\tilde{g} : Y \rightarrow X$. Then both $\tilde{g} \circ \tilde{f} : X \rightarrow X$ and $\tilde{f} \circ \tilde{g} : Y \rightarrow Y$ must be isometric embeddings. Thus, it follows that also $\tilde{f} : X \rightarrow Y$ must be bijective and therefore is an isometry. $\square$

Chapter 2

The Continuum Random Tree

2.1 Real trees

This section introduces the concept of an $\mathbb{R}$ -tree and will follow both [3], [4] and [2]. For this section, I will fix some notation. Given a metric space (X, d) , positive real numbers $a, b \in \mathbb{R}$ and continuous maps $f : [0, a] \rightarrow X$ and $g : [0, b] \rightarrow X$ with $f(a) = g(0)$ I will denote the **concatenation** f and g by the map $f * g : [0, a + b] \rightarrow X$ defined by

$$f * g(t) = \begin{cases} f(t) & , t \in [0, a] \\ g(t - a) & , t \in [a, a + b]. \end{cases}$$

By the gluing lemma¹ is again a continuous map. Also I will denote for any $s, t \in \mathbb{R}$: $s \wedge t = \min\{s, t\}$ and $s \vee t = \max\{s, t\}$.

Definition 2.1 A $\mathbb{R}$ -metric space $(\mathcal{T}, d)$ is called an **$\mathbb{R}$ -tree** if the following two properties hold for any two elements $\sigma, \sigma' \in \mathcal{T}$.

- (T1) There is a unique continuous isometry $f_{\sigma, \sigma'} : [0, d(\sigma, \sigma')] \rightarrow \mathcal{T}$ such that $f_{\sigma, \sigma'}(0) = \sigma$ and $f_{\sigma, \sigma'}(d(\sigma, \sigma')) = \sigma'$.
- (T2) If $q : [0, 1] \rightarrow \mathcal{T}$ is a continuous injection such that $q(0) = \sigma$ and $q(1) = \sigma'$ then $q([0, 1]) = f_{\sigma, \sigma'}([0, d(\sigma, \sigma')])$.

Given points $\sigma, \sigma' \in \mathcal{T}$ I denote the $[\![\sigma, \sigma']\!] := f_{\sigma, \sigma'}([0, d(\sigma, \sigma')])$ and call it **segment** from σ to σ' . A $\mathbb{R}$ -tree $(\mathcal{T}, d)$ is called a **rooted $\mathbb{R}$ -tree** if there is a distinguished element $\rho \in \mathcal{T}$, the so-called **root**. For the root $\rho \in \mathcal{T}$ we call $\sigma \in \mathcal{T}$ the segment $[\![\rho, \sigma]\!]$ the **ancestral line** of σ . For σ, σ' we write $\sigma \preceq \sigma'$ if $\sigma \in [\![\rho, \sigma']\!]$ and say σ is **ancestor** of σ' .

¹For a topological space X with a finite closed cover $(A_i)_{i \in I}$ and a family of continuous maps $\{f_i : A_i \rightarrow Y\}_{i \in I}$ such that $f_i|_{A_i \cap A_j} = f_j|_{A_i \cap A_j}$ for all $i, j \in I$ it holds that the map $f : X \rightarrow Y$ defined by $f|_{A_i} = f_i, i \in I$, defines a continuous map.

Remark 2.2 We will only be considering compact $\mathbb{R}$ -trees, i.e., whose underlying metric space is a compact metric space. This will make things easier, as in a compact metric space $(\mathcal{T}, d)$ the map $d : \mathcal{T} \times \mathcal{T} \rightarrow \mathbb{R}$ is in fact a bounded map. Therefore, this ensures that points in a compact $\mathbb{R}$ -tree will not be arbitrarily far apart. A little more informally, a compact $\mathbb{R}$ -tree $(\mathcal{T}, d)$ can be thought of as the union

$$\mathcal{T} = \bigcup_{(\sigma, \sigma') \in \mathcal{T} \times \mathcal{T}} f_{\sigma, \sigma'}([0, d(\sigma, \sigma')]),$$

which can be thought of as a very densely branched tree, where for any two points $\sigma, \sigma' \in \mathcal{T}$ we can follow the unique branch $f_{\sigma, \sigma'}([0, d(\sigma, \sigma')])$ from σ to σ' , which has finite length determined by the metric d .

I state and prove now some properties of real trees.

Proposition 2.3 Let $(\mathcal{T}, d)$ be $\mathbb{R}$ -tree. Then the, following properties hold.

- (i) If $\sigma, \sigma' \in \mathcal{T}$ and $\eta, \zeta \in \llbracket \sigma, \sigma' \rrbracket$, then $\llbracket \eta, \zeta \rrbracket \subseteq \llbracket \sigma, \sigma' \rrbracket$ and for any $t \in [0, d(\eta, \zeta)]$

$$f_{\eta, \zeta}(t) = f_{\sigma, \sigma'}(d(\sigma, \eta) + t).$$

- (ii) For any $\sigma, \sigma' \in \mathcal{T}$ it holds that

$$\llbracket \sigma, \sigma' \rrbracket = \{\eta \in \mathcal{T} \mid d(\sigma, \sigma') = d(\sigma, \eta) + d(\eta, \sigma')\}.$$

- (iii) If $\sigma, \sigma' \in \mathcal{T}$ and $\eta, \zeta \in \llbracket \sigma, \sigma' \rrbracket$ with $\eta \in \llbracket \sigma, \zeta \rrbracket$, then $\zeta \in \llbracket \eta, \sigma' \rrbracket$.

- (iv) For any $\sigma, \sigma', \sigma'' \in \mathcal{T}$ define

$$\Delta(\sigma, \sigma', \sigma'') := \max\{t \in [0, d(\sigma, \sigma') \wedge d(\sigma, \sigma'')] \mid f_{\sigma, \sigma'}(t) = f_{\sigma, \sigma''}(t)\}$$

$$\gamma(\sigma, \sigma', \sigma'') = f_{\sigma, \sigma'}(\Delta(\sigma, \sigma', \sigma'')) = f_{\sigma, \sigma''}(\Delta(\sigma, \sigma', \sigma'')) \in \llbracket \sigma, \sigma' \rrbracket \cap \llbracket \sigma, \sigma'' \rrbracket.$$

Then, it holds that

$$\{t \in [0, d(\sigma, \sigma') \wedge d(\sigma, \sigma'')] \mid f_{\sigma, \sigma'}(t) = f_{\sigma, \sigma''}(t)\} = [0, \Delta(\sigma, \sigma', \sigma'')]$$

and

$$\llbracket \sigma, \sigma' \rrbracket \cap \llbracket \sigma, \sigma'' \rrbracket = \llbracket \sigma, \gamma(\sigma, \sigma', \sigma'') \rrbracket.$$

- (v) If $\mathcal{T}$ is rooted with root $\rho \in \mathcal{T}$, then we find that for any $\sigma, \sigma' \in \mathcal{T}$

$$\sigma \preceq \sigma' \iff d(\sigma, \sigma') = d(\rho, \sigma') - d(\rho, \sigma).$$

Proof (i) Without loss of generality, assume that $d(\sigma, \eta) \leq d(\sigma, \zeta)$ (the argument is symmetric). Since $f_{\sigma, \sigma'}$ is an isometry with $f_{\sigma, \sigma'}(0) = \sigma$, we find that $\eta = f_{\sigma, \sigma'}(d(\sigma, \eta))$ and $\zeta = f_{\sigma, \sigma'}(d(\sigma, \zeta))$. Note that $d(\eta, \zeta) = |d(\sigma, \eta) - d(\sigma, \zeta)| = d(\sigma, \zeta) - d(\sigma, \eta)$. Now define for $t \in [0, d(\eta, \zeta)]$

$$\varphi(t) := f_{\sigma, \sigma'}(d(\sigma, \eta) + t).$$

Therefore, φ is an isometry with $\varphi(0) = \eta$ and $\varphi(d(\eta, \zeta)) = \zeta$, so $\varphi = f_{\eta, \zeta}$. The result follows.

- (ii) If $\eta \in \llbracket \sigma, \sigma' \rrbracket$ then $\eta = f_{\sigma, \sigma'}(t)$ for some $t \in [0, d(\sigma, \sigma')]$. Since $f_{\sigma, \sigma'}$ is an isometry, we have $d(\sigma, \eta) = t$ and $d(\eta, \sigma') = d(\sigma, \sigma') - t$ and hence

$$d(\sigma, \eta) + d(\eta, \sigma') = t + d(\sigma, \sigma') - t = d(\sigma, \sigma').$$

Conversely, suppose that $\eta \in \mathcal{T}$ such that $d(\sigma, \eta) + d(\eta, \sigma') = d(\sigma, \sigma')$. Now define $\varphi : [0, d(\sigma, \sigma')] \rightarrow \mathcal{T}$ by $\varphi = f_{\sigma, \eta} * f_{\eta, \sigma'}$. Then φ is a continuous map with $\varphi(0) = \sigma$ and $\varphi(d(\sigma, \sigma')) = \sigma'$. Moreover, φ is an isometry if restricted to $[0, d(\sigma, \eta)]$ and $[d(\sigma, \eta), d(\sigma, \sigma')]$. Using the triangle inequality for $s \in [0, d(\sigma, \eta)]$ and $t \in [d(\sigma, \eta), d(\sigma, \sigma')]$ it follows that

$$\begin{aligned} d(\varphi(s), \varphi(t)) &\leq d(\varphi(s), \varphi(d(\sigma, \eta))) + d(\varphi(d(\sigma, \eta)), \varphi(t)) \\ &= (d(\sigma, \eta) - s) + (t - d(\sigma, \eta)) \\ &= t - s. \end{aligned}$$

Assume now that in the above, we have strict inequality $d(\varphi(s), \varphi(t)) < t - s$ for some $s \leq d(\sigma, \sigma')$. Then, we would have, again employing the triangle inequality:

$$\begin{aligned} d(\sigma, \sigma') &= d(\sigma, \varphi(s)) + d(\varphi(s), \varphi(t)) + d(\varphi(t), \sigma') \\ &= s + d(\varphi(s), \varphi(t)) + d(\sigma, \sigma') - t \\ &< s + (t - s) + d(\sigma, \sigma') - t \\ &= d(\sigma, \sigma'). \end{aligned}$$

This is a contradiction; therefore it follows that φ is an isometry and by uniqueness (T1) we must have $\varphi = f_{\sigma, \sigma'}$. Thus, $\eta f_{\sigma, \sigma'}(d(\sigma, \eta)) \in \llbracket \sigma, \sigma' \rrbracket$.

- (iii) Since $\eta \in \llbracket \sigma, \zeta \rrbracket$, we must that $d(\sigma, \eta) \leq d(\sigma, \zeta)$. Thus, by (ii), it follows that

$$d(\sigma', \zeta) = d(\sigma, \sigma') - d(\sigma, \zeta) \leq d(\sigma, \sigma') - d(\sigma, \eta) = d(\sigma, \eta).$$

Thus, by (i), we must have

$$\zeta = f_{\sigma', \sigma}(d(\sigma', \zeta)) = f_{\sigma', \eta}(d(\sigma', \zeta)) \in \llbracket \sigma', \eta \rrbracket.$$

- (iv) Define

$$J = \{t \in [0, d(\sigma, \sigma') \wedge d(\sigma, \sigma'')] \mid f_{\sigma, \sigma'}(t) = f_{\sigma, \sigma''}(t)\}.$$

Then, we have $\Delta(\sigma, \sigma', \sigma'') = \max J$. If $t \in J$, let $\eta := f_{\sigma, \sigma'}(t) \in \llbracket \sigma, \sigma' \rrbracket \cap \llbracket \sigma, \sigma'' \rrbracket$, then by (i) it follows that $\llbracket \sigma, \eta \rrbracket \subseteq \llbracket \sigma, \sigma' \rrbracket \cap \llbracket \sigma, \sigma'' \rrbracket$. For $0 \leq s \leq t$ we deduce

$$f_{\sigma, \sigma'}(s) = f_{\sigma, \eta}(s - d(\sigma, \eta)) = f_{\sigma, \sigma''}(s).$$

That is $[0, t] \subseteq J$ for all $t \in J$. Therefore, J must be an interval and since it is closed, it must contain its maximum $\Delta(\sigma, \sigma', \sigma'')$. Moreover, we have that

$$\begin{aligned} \llbracket \sigma, \sigma' \rrbracket \cap \llbracket \sigma, \sigma'' \rrbracket &= \{f_{\sigma, \sigma'}(t) | t \in J\} \\ &= \llbracket \sigma, \gamma(\sigma, \sigma', \sigma'') \rrbracket. \end{aligned}$$

(v) We have the following chain of equivalences for $\sigma, \sigma' \in \mathcal{T}$:

$$\begin{aligned} \sigma \preceq \sigma' &\iff \sigma \in \llbracket \rho, \sigma' \rrbracket \\ &\stackrel{(ii)}{\iff} d(\rho, \sigma') - d(\rho, \sigma) = d(\sigma, \sigma') \end{aligned}$$

This concludes the proof of the properties of real trees. $\square$

Remark 2.4 Note that the proof of Proposition 2.3 only used Property (T1) of Definition 2.1. Therefore, it holds for any metric space satisfying (T1).

From Proposition 2.3 we get one more definition.

Definition 2.5 Let $(\mathcal{T}, d)$ be a rooted $\mathbb{R}$ -tree with root $\rho \in \mathcal{T}$. Then by Proposition 2.3 (iv), for any $\sigma, \sigma' \in \mathcal{T}$ there is a unique $\eta = \gamma\{\rho, \sigma, \sigma'\} \in \mathcal{T}$ such that $\llbracket \rho, \sigma \rrbracket \cap \llbracket \rho, \sigma' \rrbracket = \llbracket \rho, \eta \rrbracket$. We write $\eta = \sigma \wedge \sigma'$ and call η the **most recent common ancestor** of σ and σ' .

Remark 2.6 Definition 2.5 can be thought of as the minimum function on the partially ordered set $(\mathcal{T}, \preceq)$, in analogy to the minimum function on e.g. the natural numbers $\mathbb{N}$ or the real numbers $\mathbb{R}$.

2.1.1 Construction of $\mathbb{R}$ -trees

In this subsection, I will describe how to construct an $\mathbb{R}$ -tree from a given nonnegative continuous function. In fact, I will describe a continuous and Borel measurable map from the space of continuous nonnegative maps $[0, 1] \rightarrow \mathbb{R}$ vanishing at 0 and 1 to the space of compact rooted $\mathbb{R}$ -trees.

Construction 2.7 Let $g : [0, 1] \rightarrow \mathbb{R}$ be a non-negative continuous map such that $g(0) = g(1) = 0$. Then, set for any $s, t \in [0, 1]$

$$m_g(s, t) := \inf_{r \in [s \wedge t, s \vee t]} g(r)$$

and

$$d_g(s, t) := g(s) + g(t) - 2m_g(s, t).$$

By definition, we have $g(s), g(t) \geq m_g(s, t)$ and so for any $s, t \in [0, 1]$ it holds that $d_g(s, t) \geq 0$. Clearly, d_g is symmetric in s and t . Now consider, without loss of

generality, $s, t, u, v \in [0, 1]$ such that $s \leq t \leq u \leq v$. Then, we have the following set of inclusions of intervals:

$$[s, t], [t, u] \subseteq [s, u] \subseteq [s, v]$$

and

$$[t, u], [u, v] \subseteq [t, v] \subseteq [s, v].$$

This in turn yields the following inequalities:

$$m_g(s, v) \leq m_g(s, u) \leq m_g(s, t), m_g(t, u)$$

and

$$m_g(s, v) \leq m_g(t, v) \leq m_g(t, u), m_g(u, v).$$

Then, it clearly follows that

$$\begin{aligned} d_g(s, t) + d_g(u, v) &= g(s) + g(t) - 2m_g(s, t) + g(u) + g(v) - 2m_g(u, v) \\ &\leq g(s) + g(u) - 2m_g(s, u) + g(t) + g(v) - 2m_g(t, v) \\ &\leq (d_g(s, u) + d_g(t, v)) \vee (d_g(t, u) + d_g(s, v)). \end{aligned}$$

This shows that for any $s, t, u, v \in [0, 1]$ the **four-point property** holds, i.e.

$$d_g(s, t) + d_g(u, v) \leq (d_g(s, u) + d_g(t, v)) \vee (d_g(t, u) + d_g(s, v)).$$

By choosing $u = v$, we obtain the triangle inequality for d_g . So d_g defines a pseudometric. By Construction 1.5, we can introduce the following equivalence relation on $[0, 1]$

$$s \sim_g t \iff d_g(s, t) = 0$$

and have that $\mathcal{T}_g = [0, 1] / \sim_g$ together with $d_g([s], [t]) = d_g(s, t)$ is a compact metric². Denote further $p_g : [0, 1] \rightarrow \mathcal{T}_g$ the canonical projection. Moreover, we call $\rho = p_g(0) = p_g(1)$ the **root** of $\mathcal{T}_g$.

The next theorem establishes that $(\mathcal{T}_g, d_g)$ actually defines a rooted $\mathbb{R}$ -tree. The proof of this result follows [3].

Theorem 2.8 *The metric space $(\mathcal{T}_g, d_g)$ is a compact $\mathbb{R}$ -tree with root $\rho = p_g(0) = p_g(1)$.*

Before I give the quite lengthy proof of Theorem 2.8, I will show in the next lemma that we can always isometrically change the root of our constructed tree.

²It is compact since $[0, 1]$ is compact.

2. THE CONTINUUM RANDOM TREE

Lemma 2.9 *Let $s_0 \in [0, 1)$. For any $r \in \mathbb{R}$ denote $\bar{r} \in [0, 1)$ the fractional part of r , i.e. $r - \bar{r} \in \mathbb{Z}$. Define now for any the continuous map $f : [0, 1] \rightarrow \mathbb{R}$ by*

$$f(s) = g(s_0) + g(\overline{s_0 + s}) - 2m_g(s_0, \overline{s_0 + s})$$

for every $s \in [0, 1]$. Then f is continuous, non-negative, and satisfies $f(0) = f(1) = 0$, such that we can define as in 2.7 the compact metric space $(\mathcal{T}_f, d_f)$. Moreover, the metric d_f is given by

$$d_f([s], [t]) = d_g(\overline{s_0 + s}, \overline{s_0 + t}),$$

and there exists a unique isometry $R : \mathcal{T}_f \rightarrow \mathcal{T}_g$ such that for every $s \in [0, 1]$ it holds that

$$R(p_f(s)) = p_g(\overline{s_0 + s}).$$

This lemma establishes that up to isometry we can choose any $s_0 \in [0, 1]$ to be the root of our tree.

Proof f is non-negative and continuous because g is nonnegative and continuous and taking fractional part is continuous. Since $\overline{s_0 + n} = s_0$ for all $n \in \mathbb{Z}$ it follows that $f(0) = f(1) = 0$. We now consider the following cases for w.l.o.g. $s \leq t \in [0, 1]$.

- (1) Assume that $s, t \in [0, 1 - s_0]$. Note that then for all $r \in [s, t]$ we have that $\overline{s_0 + r} = s_0 + r$. Here we again have two subcases:
 - (1a) First assume that $m_g(s_0, s_0 + s) \leq m_g(s_0 + s, s_0 + t)$, then we have the following inequalities for any $r \in [s, t]$

$$\begin{aligned} m_g(s_0, s_0 + t) &\leq m_g(s_0, s_0 + r) \\ &\leq m_g(s_0, s_0 + s) \\ &\leq m_g(s_0 + s, s_0 + t). \end{aligned}$$

Clearly, we have must have that $m_g(s_0 + s, s_0 + t) = m_g(s_0, s_0 + t)$ and thus $m_g(s_0, s_0 + r) = m_g(s_0, s_0 + s) = m_g(s_0, s_0 + t)$. Then it follows that:

$$\begin{aligned} m_f(s, t) &= \inf_{r \in [s, t]} (g(s_0) + g(s_0 + r) - 2m_g(s_0, s_0 + r)) \\ &= g(s_0) + m_g(s_0 + s, s_0 + t) - 2m_g(s_0, s_0 + s). \end{aligned}$$

Thus, we can compute

$$\begin{aligned} d_f([s], [t]) &= f(s) + f(t) - 2m_f(s, t) \\ &= g(s_0 + s) - 2m_g(s_0, s_0 + s) \\ &\quad + g(s_0 + t) - 2m_g(s_0, s_0 + t) \\ &\quad - 2(m_g(s_0 + s, s_0 + t) - 2m_g(s_0, s_0 + s)) \\ &= d_g([s_0 + s], [s_0 + t]). \end{aligned}$$

(1b) Now assume that $m_g(s_0, s_0 + s) > m_g(s_0 + s, s_0 + t)$. Denote now

$$r_1 := \inf\{r \in [s, t] \mid g(s_0 + r) = m_g(s_0, s_0 + s)\}.$$

First note that $m_g(s_0, s_0 + r_1) = m_g(s_0, s_0 + s)$. Then for $r \in [s, t]$, we have

$$g(s_0 + r) - 2m_g(s_0, s_0 + r) \geq -m_g(s_0, s_0 + r).$$

Thus if $r \leq r_1$ as $m_g(s_0, s_0 + r) \leq m_g(s_0, s_0 + s)$ we find

$$g(s_0 + r) - 2m_g(s_0, s_0 + r) \geq -m_g(s_0, s_0 + s).$$

Similarly, for $r \geq r_1$ we have $m_g(s_0, s_0 + r_1) \geq m_g(s_0, s_0 + r)$ and so

$$g(s_0 + r) - 2m_g(s_0, s_0 + r) \geq -m_g(s_0, s_0 + s).$$

And as

$$g(s_0 + r) - 2m_g(s_0, s_0 + r) = -m_g(s_0, s_0 + s),$$

$m_f(s, t)$ is attained at r_1 , i.e.

$$m_f(s, t) = g(s_0) - m_g(s_0, s_0 + s).$$

Then, we can compute

$$\begin{aligned} d_f([s], [t]) &= g(s_0 + s) - 2m_g(s_0, s_0 + s) \\ &\quad + g(s_0 + t) - 2m_g(s_0, s_0 + t) \\ &\quad + 2m_g(s_0, s_0 + s) \\ &= d_g([s_0 + s], [s_0 + t]). \end{aligned}$$

In the last line, we used

$$m_g(s_0, s_0 + t) \leq m_g(s_0 + s, s_0 + t) < m_g(s_0, s_0 + s),$$

and therefore $m_g(s_0, s_0 + t) = m_g(s_0 + s, s_0 + t)$.

(2) If $s \in [1 - s_0, 1]$ or $t \in [1 - s_0, 1]$ we note that

$$\overline{s_0 + s} = s_0 + s - 1 \in [0, 1 - s_0) \text{ and } \overline{s_0 + t} = s_0 + t - 1 \in [0, 1 - s_0).$$

Suppose without loss of generality that $s \in [1 - s_0, 1]$ and so $\overline{s_0 + s} = s_0 + s - 1$, then we make the same distinctions as in (1a) and (1b) up to replacing $s_0 + s$ by $s_0 + s - 1$, which then yields the same result.

Therefore, in any case, we have shown that

$$d_f([s], [t]) = d_g(\overline{[s_0 + s]}, \overline{[s_0 + t]}). \quad (*)$$

By this, we can deduce for any $s, t \in [0, 1]$, that $d_f([s], [t]) = 0$ if and only if $d_g(\overline{[s_0 + s]}, \overline{[s_0 + t]}) = 0$ and so $p_g(\overline{[s_0 + s]}) = p_g(\overline{[s_0 + t]})$. That is, the equivalence classes of $\mathcal{T}_f$ and $\mathcal{T}_g$ are in a one-to-one correspondence defined by

$$R : \mathcal{T}_f \rightarrow \mathcal{T}_g; [s] \mapsto \overline{[s_0 + s]}.$$

The map R is well-defined by the identity $(*)$, we have shown above. Moreover, by the identity $(*)$ we see immediately that R defines an isometry $\mathcal{T}_f \rightarrow \mathcal{T}_g$ and satisfies by definition for every $s \in [0, 1]$

$$R(p_f(s)) = p_g(\overline{[s_0 + s]}).$$

This concludes the proof. $\square$

Before turning to the proof of Theorem 2.8 I will make some preliminary lemmas.

Lemma 2.10 *Consider the metric space $(\mathcal{T}_g, d_g)$ with $\rho = p_g(0) = p_g(1)$. Then consider in analogy to Definition 2.1, Proposition 2.3 and Definition 2.5 the following*

- (i) *For $\sigma, \sigma' \in \mathcal{T}$ set $\sigma \preceq \sigma'$ if and only if $d_g(\sigma, \sigma') = d_g(\rho, \sigma') - d_g(\rho, \sigma)$. Then, if $\sigma = p_g(s)$ and $\sigma' = p_g(t)$ it holds that*

$$\sigma \preceq \sigma' \iff g(s) = m_g(s, t).$$

This defines a partial order³ on $\mathcal{T}_g$. In particular, if Theorem 2.8 is true $\preceq$ is exactly the ancestral relation given in Definition 2.1 by Proposition 2.3 (v).

- (ii) *In consistency with Proposition 2.3 define for $\sigma, \sigma' \in \mathcal{T}$ the set*

$$\llbracket \sigma, \sigma' \rrbracket := \{\eta \in \mathcal{T}_g \mid d_g(\sigma, \sigma') = d_g(\sigma, \eta) + d_g(\eta, \sigma')\}.$$

Now if $\sigma = p_g(s)$ and $\sigma' = p_g(t)$, then

$$\llbracket \rho, \sigma \rrbracket \cap \llbracket \rho, \sigma' \rrbracket = \llbracket \rho, \eta \rrbracket$$

where $\eta = p_g(r)$ for any $r \in [0, 1]$ such that $g(r) = m_g(s, t)$. We denote such $\eta \in \mathcal{T}$ by $\eta = \sigma \wedge \sigma'$.

- (iii) *We define the **children** of $\sigma \in \mathcal{T}_g$ by*

$$\mathcal{T}_g[\sigma] := \{\sigma' \in \mathcal{T}_g \mid \sigma \preceq \sigma'\}.$$

If $\mathcal{T}_g[\sigma] \neq \{\sigma\}$ and $\sigma \neq \rho$, then $\mathcal{T}_g \setminus \mathcal{T}_g[\sigma]$ and $\mathcal{T}_g[\sigma] \setminus \{\sigma\}$ are non-empty disjoint open sets.

³ $\preceq$ is reflexive and antisymmetric by the positive definiteness of d_g and transitive by the triangle inequality of d_g .

Proof (i) We note first that for any $r \in [0, 1]$ it holds that $m_g(r, 0) = 0$ as g is non-negative and $g(0) = 0$. So since $\rho = p_g(0)$ we find that for any $\tau = p_g(r) \in \mathcal{T}_g$ it holds that $d_g(\tau, \rho) = g(r)$. Hence, we obtain the following equivalences for $\sigma = p_g(s)$ and $\sigma' = p_g(t)$:

$$\begin{aligned} \sigma \preceq \sigma' &\iff d_g(\sigma, \sigma') = d_g(\rho, \sigma') - d_g(\rho, \sigma) \\ &\iff g(s) + g(t) - 2m_g(s, t) = g(t) - g(s) \\ &\iff 2g(s) = 2m_g(s, t) \\ &\iff g(s) = m_g(s, t) \end{aligned}$$

(ii) For this I direct the reader to (iv) in the proof of Lemma 2.3.

(iii) Let $s \in [0, 1]$ such that $\sigma = p_g(s)$ and note that $\mathcal{T}_g[\sigma]$ is the image of the compact set $\{u \in [0, 1] \mid m_g(s, u) = g(s)\}$ under the continuous map p_g . Hence, $\mathcal{T}_g \setminus \mathcal{T}_g[\sigma]$ is open. Now for $\sigma' \in \mathcal{T}_g[\sigma] \setminus \{\sigma\}$ consider the open ball $B_r(\sigma')$ with radius $r = d_g(\sigma, \sigma')$. Then it holds for all $\eta = p_g(u) \in B_r(\sigma')$ that

$$\begin{aligned} d_g(\sigma', \eta) < r &\implies g(u) + g(s) < m_g(u, t) \\ &\implies m_g(s, u) < m_g(u, t) \\ &\stackrel{\sigma \preceq \sigma'}{\implies} m_g(s, u) = m_g(s, t) = g(s) \\ &\implies \sigma \preceq \eta. \end{aligned}$$

Hence, it follows that $B_r(\sigma') \subseteq \mathcal{T}_g[\sigma] \setminus \{\sigma\}$. This shows that $\mathcal{T}_g[\sigma] \setminus \{\sigma\}$ is open. $\square$

Using Lemma 2.10 and Lemma 2.9 we are now equipped to prove Theorem 2.8.

Proof First, we prove property (T1) of Definition 2.1 of $\mathbb{R}$ -trees. Let $\sigma, \sigma' \in \mathcal{T}_g$. By Lemma 2.9, we can assume up to isometry that $\sigma' = \rho = p_g(0)$. To show is that there is a unique isometry $f : [0, d_g(\rho, \sigma)] \rightarrow \mathcal{T}_g$ such that $f(0) = \rho$ and $f(d_g(\rho, \sigma)) = \sigma$. Now, select representative $s \in p_g^{-1}(\{\sigma\})$, so that $g(s) = d_g(\rho, \sigma)$. Then for any $a \in [0, d_g(\rho, \sigma)]$ we define

$$\omega(a) := \inf\{r \in [0, s] \mid m_g(r, s) = a\}.$$

Therefore, $g(\omega(a)) = a$. Set now $f(a) := p_g(\omega(a))$. We have that $f(0) = 0$ and $f(d_g(\rho, \sigma)) = \sigma$, since $g(s) = d_g(\rho, \sigma)$ we find

$$\begin{aligned} m_g(\omega(g(s)), s) = g(s) &\implies d_g(\omega(g(s), s), s) = 0 \\ &\implies f(d_g(\rho, \sigma)) = p_g(\omega(g(s))) = p_g(s) = \sigma \end{aligned}$$

We now verify that f is an isometry: Suppose $a, b \in [0, d_g(\rho, \sigma)]$ with $a \leq b$, then we obtain

$$m_g(\omega(a), \omega(b)) = a,$$

and so we get

$$d_g(f(a), f(b)) = g(\omega(a)) + g(\omega(b)) - 2a = a + b - 2a = b - a.$$

This proves that f is an isometry. Now, we turn our attention to the uniqueness of f . Suppose $\tilde{f} : [0, d_g(\rho, \sigma)] \rightarrow \mathcal{T}_g$ another isometry such that $\tilde{f}(0) = \rho$ and $\tilde{f}(d_g(\rho, \sigma)) = \sigma$. Then, for any $a \in [0, d_g(\rho, \sigma)]$ we find

$$\begin{aligned} d_g(\tilde{f}(a), \sigma) &= d_g(\rho, \sigma) - a \\ &= d_g(\rho, \sigma) - d_g(\rho, \tilde{f}(a)). \end{aligned}$$

Hence, by the Lemma 2.10 we have that $\tilde{f}(a) \preceq \sigma$. We have $\sigma = p_g(s)$ and $\tilde{f}(a) = p_g(t)$ for some $t \in [0, 1]$, then by Lemma 2.10 (i) we have that $g(t) = m_g(t, s)$ and $g(t) = d_g(\rho, p_g(t)) = a$. Then we have that

$$a = g(t) = g(\omega(a)) = m_g(\omega(a), t).$$

Thus, it follows that

$$\begin{aligned} d_g(t, \omega(a)) &= g(t) + g(\omega(a)) - 2m_g(t, \omega(a)) \\ &= a + a - 2a \\ &= 0. \end{aligned}$$

Therefore, it must follow that $\tilde{f}(a) = p_g(t) = p_g(\omega(a)) = f(a)$. Hence, $\tilde{f} = f$. We have already seen in the argument above for uniqueness that for any $a \in [0, d_g(\rho, \sigma)]$ we have that $f(a) \preceq \sigma$. Hence, $f([0, d_g(\rho, \sigma)]) \subseteq \llbracket \rho, \sigma \rrbracket$. On the other hand, if $\eta \preceq \sigma$, then by the argument above we get $\eta = f(d_g(\rho, \eta))$. Hence, $f([0, d_g(\rho, \sigma)]) = \llbracket \rho, \sigma \rrbracket$.

Now, we prove the property (T2) of Definition 2.1. Fix a continuous injection $q : [0, 1] \rightarrow \mathcal{T}_g$. Up to isometry by Lemma 2.9 we can assume that $q(0) = \rho$ and $q(1) = \sigma$. From the first part of the proof, we have $f_{\rho, \sigma}([0, d_g(\rho, \sigma)]) = \llbracket \rho, \sigma \rrbracket$. Suppose now there is $\eta \in \llbracket \rho, \sigma \rrbracket \setminus q([0, 1])$. In particular, we have that $\eta \neq \rho, \sigma$. Now consider the non-empty disjoint open sets $A = \mathcal{T}_g \setminus \mathcal{T}_g[\eta]$ and $B = \mathcal{T}_g[\eta] \setminus \{\eta\}$. Then $q([0, 1]) \subseteq A \cup B$ and $q(0) = \rho \in A$ and $q(1) = \sigma \in B$. This is a contradiction, because $q([0, 1])$ is connected. This shows that $q([0, 1]) \subseteq \llbracket \rho, \sigma \rrbracket$. Conversely, suppose there is $a \in (0, 1)$ such that $q(a) \notin \llbracket \rho, \sigma \rrbracket$. Set $\eta = q(a)$ and choose $\zeta = \sigma \wedge \eta$. By definition, we have $\zeta = \sigma \wedge \eta \in \llbracket \rho, \eta \rrbracket \cap \llbracket \eta, \sigma \rrbracket$ and hence, from the first part of the proof of (T2), we must have $\zeta \in q([0, a])$ as $\zeta \in \llbracket \rho, \eta \rrbracket$ and up to changing the root ρ with σ we also $\zeta \in q([a, 1])$. Then, we would have $\zeta = q(a) = \eta$. $\square$

This shows that $(\mathcal{T}_g, d_g)$ is a compact rooted $\mathbb{R}$ -tree with root $\rho = p_g(0) = p_g(1)$ for any continuous non-negative map $g : [0, 1] \rightarrow \mathbb{R}$ such that $g(0) = g(1) = 0$. Define now $\mathbb{T}$ to be the space of the isometry classes of compact rooted $\mathbb{R}$ -trees. This is a metric subspace of the space $\mathbb{K}$ of isometry classes

of compact metric spaces and is therefore equipped with Gromov-Hausdorff distance d_{GH} . We recall that for compact metric spaces (X, d_X) and (Y, d_Y) we have that

$$d_{GH}(X, Y) = \frac{1}{2} \inf_{\mathcal{R}} \text{dis} \mathcal{R},$$

by Proposition 1.12, where the infimum is taken over the set of bisurjective relations $\mathcal{R}$ on $X \times Y$ and $\text{dis} \mathcal{R}$ is the distortion of $\mathcal{R}$, i.e.

$$\text{dis} \mathcal{R} := \sup \{ |d_X(x, x') - d_Y(y, y')| \mid (x, y), (x', y') \in \mathcal{R} \}.$$

Denote in the following $(\mathcal{C}([0, 1]), d_\infty)$ the metric space of non-negative continuous functions $g : [0, 1] \rightarrow \mathbb{R}$ such that $g(0) = g(1) = 0$ together with the supremum metric given by

$$d_\infty(f, g) = \sup_{t \in [0, 1]} |f(t) - g(t)|.$$

In the next theorem, we will consider the following map:

$$\mathcal{T}_\bullet : \mathcal{C}([0, 1]) \rightarrow \mathbb{T}; g \mapsto \mathcal{T}_g.$$

This is clearly a well-defined map, since $\mathcal{T}_g$ is an $\mathbb{R}$ -tree by Theorem 2.8. The next theorem establishes that the map $\mathcal{T}_\bullet$ is a continuous map and therefore, in particular, Borel measurable.

Theorem 2.11 *For all $f, g \in \mathcal{C}([0, 1])$ it holds that*

$$d_{GH}(\mathcal{T}_f, \mathcal{T}_g) \leq 2d_\infty(f, g).$$

Proof We construct the following bisurjective relation between $\mathcal{T}_f$ and $\mathcal{T}_g$:

$$\mathcal{R} = \{(\sigma, \sigma') \in \mathcal{T}_f \times \mathcal{T}_g \mid \exists t \in [0, 1] : \sigma = p_f(t), \sigma' = p_g(t)\}.$$

One can easily see that this is actually a bisurjective relation: Since $p_f : [0, 1] \rightarrow \mathcal{T}_f$ is surjective for any $\sigma \in \mathcal{T}_f$ we find $t \in [0, 1]$ such that $\sigma = p_f(t)$. Since $p_g : [0, 1] \rightarrow \mathcal{T}_g$ is a function, we have $\sigma' = p_g(t) \in \mathcal{T}_g$ such that $(\sigma, \sigma') \in \mathcal{R}$. By the same reasoning, for any $\sigma' \in \mathcal{T}_g$ we find $\sigma \in \mathcal{T}_f$ such that $(\sigma, \sigma') \in \mathcal{R}$.

Fix $(\sigma, \sigma'), (\eta, \eta') \in \mathcal{R}$. By definition of $\mathcal{R}$ there are some $s, t \in [0, 1]$ such that $p_f(s) = \sigma$, $p_g(s) = \sigma'$, $p_f(t) = \eta$ and $p_g(t) = \eta'$. Then we have that

$$d_f(\sigma, \eta) = f(s) + f(t) - 2m_f(s, t)$$

and

$$d_g(\sigma', \eta') = g(s) + g(t) - 2m_g(s, t).$$

Now, let $r, r' \in [s \wedge t, s \vee t]$ such that

$$f(r) = m_f(s, t) \text{ and } g(r') = m_g(s, t).$$

Then for $u \in \{r, r'\}$ we find

$$\begin{aligned} |d_f(\sigma, \eta) - d_g(\sigma', \eta')| &= |(f(s) - g(s)) + (f(t) - g(t)) + 2(g(r') - f(r))| \\ &\leq |f(s) - g(s)| + |f(t) - g(t)| + 2|g(u) - f(u)| \\ &\leq 4d_\infty(f, g). \end{aligned}$$

This shows that $\text{dis}\mathcal{R} \leq 4d_\infty(f, g)$. By Proposition 1.12 we then find that

$$d_{GH}(\mathcal{T}_f, \mathcal{T}_g) = \frac{1}{2} \inf_{\mathcal{R}} \text{dis}\mathcal{R} \leq 2d_\infty(f, g).$$

Therefore, the map $\mathcal{T}_\bullet$ is 2-Lipschitz continuous and in particular continuous. $\square$

This concludes the construction of the coding of $\mathbb{R}$ -trees.

2.2 Stochastic processes, Brownian motion, and Brownian excursion

Because the map

$$g \in \mathcal{C}([0, 1]) \mapsto \mathcal{T}_g \in \mathbb{T}$$

is a continuous map by Theorem 2.11, it is particularly Borel measurable with respect to Borel σ -algebras on the metric spaces $(\mathcal{C}([0, 1]), d_\infty)$ and $(\mathbb{T}, d_{GH})$. This enables us to introduce probability into the equation. That is, we can consider a *continuous stochastic process* $X = (X_t)_{t \in [0, 1]}$ with $X_0 = X_1 = 0$ *almost surely* and consider the *random tree* $\mathcal{T}_X$. In this section, we introduce the theory of stochastic processes used for this, starting with the definition of a stochastic process. We will follow Chapters I of [5]. This section will lack proofs, as this will go beyond the focus of this thesis. It should mainly provide the ideas and definitions of the probability theory used in this thesis. The constructions given in this section should be taken with a grain of salt as they should all be treated with care in their own right (where I direct the reader to [5]).

Definition 2.12 *Let (E, d) be a metric space. A **continuous stochastic process** with values in E for us will be a family $X = \{X_t\}_{t \geq 0}$ of measurable maps X_t from a probability space $(\Omega, \mathcal{F}, P)$ to $(E, \mathcal{B}(E))$, where $\mathcal{B}(E)$ is the Borel σ -algebra on E , such that for almost every⁴ $\omega \in \Omega$ the map*

$$X_\bullet(\omega) : t \in [0, \infty) \mapsto X_t(\omega) \in E$$

⁴We say a property A holds for **almost every** $\omega \in \Omega$ if

$$P(\{\omega \in \Omega \mid A \text{ does not hold for } \omega\}) = 0.$$

is continuous. We may consider a continuous stochastic process X also as a measurable map

$$X : (\Omega, \mathcal{F}, P) \rightarrow (C^0([0, \infty), E), \mathcal{B}(C^0([0, \infty), E))); \omega \mapsto X_\bullet(\omega),$$

where $C^0([0, \infty), E)$ is the metric space of continuous maps $[0, \infty) \rightarrow E$ (with supremum metric d_∞) and $\mathcal{B}(C^0([0, \infty), E))$ the respective Borel σ -algebra.

Notation 2.13 For the remainder of this chapter, we fix a probability space $(\Omega, \mathcal{F}, P)$ together with a filtration $\mathbb{F}$ of the σ -algebra $\mathcal{F}$. I will refer to real-valued measurable maps defined on $(\Omega, \mathcal{F}, P)$ as **random variables**. To measurable maps on $(\Omega, \mathcal{F}, P)$ with values in a metric space (E, d) (or in fact any measurable space), I will refer to as **E-valued random variables**.

Equipped with this definition, we may define the most important continuous stochastic process, the so-called *Brownian motion*. First, we define the notion of a *Gaussian random variable and process*.

Definition 2.14 A **Gaussian random variable** is a (continuous) random variable $Y : (\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that there exists $\mu \in \mathbb{R}$ and $\sigma > 0$ and Y has the probability density

$$\varphi_{\mu, \sigma}(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right).$$

We now write $X \sim \mathcal{N}(\mu, \sigma^2)$. One calls μ its **mean** and σ resp. σ^2 its **standard deviation** resp. its **variance**. A **Gaussian process** is a continuous stochastic process $X = \{X_t\}_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ such that

$$\forall n \in \mathbb{N} \forall 0 \leq t_1 \leq \dots \leq t_n \forall (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n : \\ \sum_{i=1}^n \lambda_i X_{t_i} \text{ is a Gaussian random variable.}$$

Now, *Brownian motion* is a special stochastic process, whose increments are independent⁵ Gaussian random variables.

Definition 2.15 A **(1-dimensional) Brownian motion** is a continuous $\mathbb{R}$ -valued process $B = \{B_t\}_{t \geq 0}$ such that these properties are satisfied:

⁵Recall for metric space (E, d) that two E -valued random variables Y and Z on $(\Omega, \mathcal{F}, P)$ are called independent if the σ -algebras $\sigma(Y)$ and $\sigma(Z)$ generated by Y and Z (i.e. the smallest σ -algebras for which Y resp. Z are measurable) are independent, that is

$$\forall (A, B) \in \sigma(Y) \times \sigma(Z) : P(A \cap B) = P(A)P(B).$$

2. THE CONTINUUM RANDOM TREE

(B1) B has independent increments, i.e., for any $n \in \mathbb{N}$ and sequence $t_0 = 0 < t_1 < \dots < t_n$ the random variables $B_{t_i} - B_{t_{i-1}}$, $i = 1, \dots, n$, are independent⁵.

(B2) For all $0 \leq s < t$ the random variables $B_t - B_s$ is a Gaussian random variables with mean 0 and variance $t - s$, i.e. $B_t - B_s \sim \mathcal{N}(0, t - s)$.

One might ask, is this definition justified? Does such a stochastic process even exist? This is in fact a valid question. However, as the answer warrants an exposition on its own and is only tangentially of interest in this thesis, I will only give the idea of the construction direct the interested reader to [6].

Construction 2.16 Here, I will give the construction I will give the construction using the Daniell-Kolmogorov theorem (1919) as adapted from [6]: Consider a family $\{(\Omega_i, \mathcal{F}_i)\}_{i \in I}$ of measurable spaces and the set $\mathcal{E}(I)$ of finite subsets of I and for any finite subset $J \subseteq I$ a probability measure P_J on the measurable space

$$(\Omega^J, \mathcal{F}^J) := \left(\prod_{j \in J} \Omega_j, \bigotimes_{j \in J} \mathcal{F}_j \right),$$

such that for the projections $\pi_K^J : \Omega^J \rightarrow \Omega^K$ for $K \subseteq J$ it holds that for any $K \subseteq J$:

$$Q_K = Q_J \circ (\pi_K^J)^{-1}.$$

Then, the family $\{P_J\}_{J \in \mathcal{E}(I)}$ is called **consistent**. The Daniell-Kolmogorov theorem states that

Theorem 2.17 (Daniell-Kolmogorov) If $\{Q_J\}_{J \in \mathcal{E}(I)}$ is a consistent family of probability measures, then there exists a unique probability measure P on the measurable space

$$(\Omega, \mathcal{F}) = \left(\prod_{i \in I} \Omega_i, \bigotimes_{i \in I} \mathcal{F}_i \right)$$

such that for any $J \in \mathcal{E}(I)$ it holds that

$$P_J = P \circ (\pi_J^I)^{-1}.$$

To construct Brownian motion, we use this theorem and show that on the measurable space

$$(\Omega, \mathcal{F}) = (\mathbb{R}^{[0, \infty)}, \mathcal{B}(\mathbb{R}^{[0, \infty)})) = \left(\prod_{t \in [0, \infty) \mathbb{R}}, \bigotimes_{t \in [0, \infty)} \mathcal{B}(\mathbb{R}) \right)$$

there exists a unique probability measure P such that the canonical process B defined by

$$B_t(\omega) = \omega(t) = \pi_{\{t\}}^{[0, \infty)}(\omega), \quad \omega \in \Omega, t \geq 0$$

2.2. Stochastic processes, Brownian motion, and Brownian excursion

is Brownian motion according to Definition 2.15. For this, let $I = [0, \infty)$ and define for any finite subset $J = \{0 = t_0 < t_1 < \dots < t_n\} \subseteq I$ the probability measure P_J on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ by

$$P_J(A) = \int_A \prod_{i=1}^n \varphi_{0, t_i - t_{i-1}}(x_i) d\mu(x), \quad A \in \mathcal{B}(\mathbb{R}^n),$$

where $\varphi_{0, t_i - t_{i-1}}$ is probability density of $\mathcal{N}(0, t_i, t_{i-1})$ and μ the Lebesgue measure on $\mathbb{R}^n$. One can check that the family $\{P_J\}_{J \in \mathcal{E}(I)}$ is consistent, so by Theorem 2.17 there exists a probability measure P on $(\Omega, \mathcal{F})$, such that $P_J = P \circ (\pi_J^I)^{-1}$. In particular, for any $t \geq 0$ and $A \in \mathcal{B}(\mathbb{R})$ we have that

$$P(B_t \in A) = P \circ \left(\pi_{\{t\}}^{[0, \infty)} \right)^{-1} (A) = P_{\{t\}}(A),$$

that is $B_t \sim \mathcal{N}(0, t)$. One can now show that the process B together with the probability measure satisfies (B1) and (B2) of Definition 2.15 and is (up to showing continuity) a Brownian motion.

This concludes the discussion about Brownian motion. To continue on to the construction and properties of the *continuum random tree* one key piece of probability theory, which remains to be introduced, is the *excursion process* of Brownian motion. Here I will give the construction and definition given in [7], which I notationally slightly adapted.

Construction 2.18 Let B be a Brownian motion and define for the stopping times⁶

$$\begin{aligned} \tau_-(\omega) &= \sup\{t \in [0, 1] \mid B_t(\omega) = 0\} \\ \tau_+(\omega) &= \inf\{t \geq 1 \mid B_t(\omega) = 0\}. \end{aligned}$$

Then, we defines the process $e = \{e_t\}_{t \in [0, 1]}$ by

$$e_t(\omega) := (\tau_+ - \tau_-)^{-\frac{1}{2}} \left| B_{(1-t)\tau_-(\omega) + t\tau_+(\omega)}(\omega) \right|.$$

This can be shown to be the Brownian motion B conditioned to be positive on $[0, 1]$ (for this consult the theory explained in Chapter XII of [5] on excursion processes).

Construction 2.18 leads to the following definition.

Definition 2.19 For a Brownian motion B , the continuous process $e = \{e_t\}_{t \in [0, 1]}$ defined in Construction 2.18 is called **Brownian excursion**.

⁶A stopping time τ is a $[0, \infty)$ -valued random variable on $(\Omega, \mathcal{F}, P)$ with filtration $\mathbb{F}$ such that for any $t \geq 0$ it holds that

$$\{\tau \leq t\} = \{\omega \in \Omega \mid \tau(\omega) \leq t\} \in \mathcal{F}_t.$$

Remark 2.20 Note that as eluded in Definition 2.12 a Brownian excursion e can be seen as a measurable map

$$e : \omega \in (\Omega, \mathcal{F}, P) \mapsto e_\bullet(\omega) \in (C^0([0, 1]), \mathcal{B}(C^0([0, 1]))).$$

Moreover, we can check that almost surely $e_0 = e_1 = 0$ and hence e can even be seen as a measurable map

$$e : \omega \in (\Omega, \mathcal{F}, P) \mapsto e_\bullet(\omega) \in (\mathcal{C}([0, 1]), \mathcal{B}(\mathcal{C}([0, 1]))).$$

This concludes these probability theoretic preliminaries, and we can now consider the construction of the *continuum random tree*.

2.3 Definition and properties of the continuum random tree

In this section, I will discuss the definition and properties of the so-called *continuum random tree*. For this fix a Brownian excursion

$$e : \omega \in (\Omega, \mathcal{F}, P) \mapsto e(\omega) = e_\bullet(\omega) \in (\mathcal{C}([0, 1]), \mathcal{B}(\mathcal{C}([0, 1]))).$$

seen as a measurable map on the probability space $(\Omega, \mathcal{F}, P)$ with values in the metric space $(\mathcal{C}([0, 1]), d_\infty)$ of non-negative continuous maps $[0, 1] \rightarrow \mathbb{R}$ vanishing at 0 and 1 (see Remark 2.20). Recall the continuous map (see Construction 2.7 and Theorems 2.8 and 2.11)

$$\mathcal{T}_\bullet : \mathcal{C}([0, 1]) \rightarrow \mathbb{T}$$

which codes compact rooted $\mathbb{R}$ -trees.

Definition 2.21 The *continuum random tree (CRT)* is the continuous $\mathbb{T}$ -valued random variable

$$\mathcal{T}_e : \omega \in (\Omega, \mathcal{F}, P) \mapsto (\mathcal{T}_{e(\omega)}, d_{GH}) \in (\mathbb{T}, \mathcal{B}(\mathbb{T})).$$

That is, one can see $(\mathcal{T}_e, d_{GH})$ as a random compact rooted $\mathbb{R}$ -tree.

2.3.1 Discrete trees and CRT as the limit of Galton-Watson trees

This will be the first property of the CRT, which characterizes it as the *distributional limit* of so-called *Galton-Watson trees*, which will be defined using the reference [8]. First, we need the concept of a *discrete tree*.

Definition 2.22 Define the *set of labels*

$$\mathcal{U} = \bigcup_{n=0}^{\infty} \mathbb{N}^n.$$

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with the convention $\mathbb{N}^0 = \{\emptyset\}$. An element $u = (u^1, \dots, u^n) \in \mathbb{N}^n \subset \mathcal{U}$ is called a **label** of **generation** $|u| := n$. Given two labels $u = (u^1, \dots, u^n)$ and $v = (v^1, \dots, v^m)$ define their **concatenation** by

$$\overline{uv} := (u^1, \dots, u^n, v^1, \dots, v^m) = (u, v),$$

with the convention $\overline{u\emptyset} = \overline{\emptyset u} = u$. Additionally, define the mapping $\pi : \mathcal{U} \setminus \{\emptyset\} \rightarrow \mathcal{U}$ by

$$\pi(u^1, \dots, u^n) = (u^1, \dots, u^{n-1}), \quad n \in \mathbb{N}.$$

A **finite rooted ordered tree** is a finite subset $\mathbf{t} \subset \mathcal{U}$ such that the following properties hold:

- (t1) $\emptyset \in \mathbf{t}$.
- (t2) $u \in \mathbf{t} \setminus \{\emptyset\} \implies \pi(u) \in \mathbf{t}$.
- (t3) For every $u = (u^1, \dots, u^n) \in \mathbf{t}$ there exists integer $\kappa_u(\mathbf{t}) \geq 0$ such that for any $j \in \mathbb{N}$ it holds that

$$\overline{uj} = (u^1, \dots, u^n, j) \in \mathbf{t} \iff 1 \leq j \leq \kappa_u(\mathbf{t}).$$

Let $\mathbf{A}$ denote the set of all finite rooted ordered trees and denote by $\#(\mathbf{t})$ the cardinality of $\mathbf{t}$. In addition, if $\kappa_u(\mathbf{t}) = 0$ for some $u \in \mathbf{t}$ we call u a **leaf** of $\mathbf{t}$.

I will describe two important functions associated with a finite rooted ordered tree.

Construction 2.23 Let $\mathbf{t}$ be a finite rooted ordered tree. Then, we may order the elements of $\mathbf{t}$ in lexicographical order⁷:

$$u(0) = \emptyset, u(1), \dots, u(\#(\mathbf{t}) - 1) \in \mathbf{t}.$$

The **height function** associated with $\mathbf{t}$ is given by

$$h(\mathbf{t}) : \{0, \dots, \#(\mathbf{t}) - 1\} \rightarrow \mathbb{N}; n \mapsto h_n(\mathbf{t}) := |u(n)|.$$

The **contour function** associated with $\mathbf{t}$ is defined as follows:

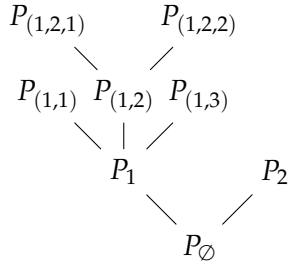
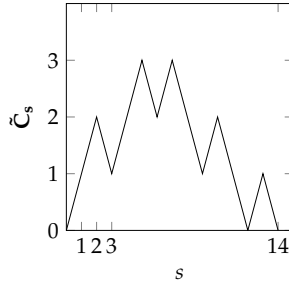
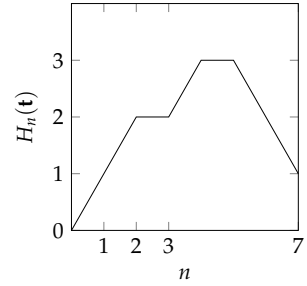
Suppose the tree $\mathbf{t}$ is embedded in the plane $\mathbb{R}^2$ by associating to each $u \in \mathbf{t}$ a point $P_u \in \mathbb{R}^2$ in such a way that for all $1 \leq j \leq \kappa_u(\mathbf{t})$ there is a line of length 1 (in Euclidean metric) for P_u to $P_{\overline{uj}}$. By convention, we set $P_\emptyset = (0, 0)$. Suppose now that there is a particle $C(\mathbf{t})$ starting at $P_\emptyset = (0, 0)$ at time $s = 0$ moving on the

⁷For $u, v \in \mathbf{t}$ that means $(u^1, \dots, u^m) \leq_{\text{lex}} (v^1, \dots, v^n)$ if and only if

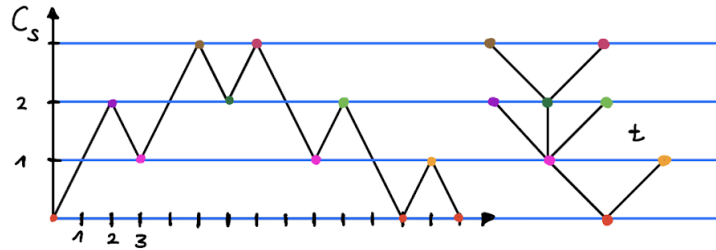
$(\exists k \leq \min\{m, n\} : (\forall j < k : u^j = v^j) \text{ and } u^k \leq v^k) \text{ or } (m \leq n \text{ and } \forall j \leq m : u^j = v^j).$

lines drawn between the points P_u at unit speed in the following way: It moves from the left to right, visiting at any P_u the points $P_{u\bar{j}}$ for $1 \leq j \leq \kappa_u(\mathbf{t})$. If it hits $P_{u\bar{j}}$ with $\kappa_{u\bar{j}}(\mathbf{t}) = 0$ then it will move back along the line between P_u and $P_{u\bar{j}}$ to P_u and continue to $P_{u\bar{j+1}}$. It will stop when it has visited every line twice, i.e., at time $2(\#\mathbf{t}) - 1$ again. At each time $s \in [0, 2(\#\mathbf{t}) - 1]$ we record the distance $\tilde{C}_s(\mathbf{t})$ of the particle to the point $P_\emptyset = (0, 0)$. We now define the **contour function associated to $\mathbf{t}$** by

$$C(\mathbf{t}) : [0, 1] \rightarrow \mathbb{R}; s \mapsto \tilde{C}_{(2(\#\mathbf{t})-1)s}(\mathbf{t}).$$


 Figure 2.1: tree $\mathbf{t}$

 Figure 2.2: contour function $C(\mathbf{t})$

 Figure 2.3: height function $H(\mathbf{t})$

Remark 2.24 One can view any finite rooted ordered tree $\mathbf{t} \in \mathbf{A}$ as a rooted $\mathbb{R}$ -tree, by embedding it into the plane as seen in Construction 2.23 in Figure 2.1, that is, with any $u \in \mathcal{T}$ one associates a point $P_u \in \mathbb{R}^2$ with the convention $P_\emptyset = (0, 0)$. For any $j \in \{1, \dots, \kappa_u(\mathbf{t})\}$ we then draw a line segment of length 1 from P_u to $P_{u\bar{j}}$. We let can view then $\mathbf{t}$ as the union of these line segments. Then, we obtain something as seen in Figure 2.1. We can equip then this embedding, also denoted $\mathbf{t}$, of the tree $\mathbf{t}$ with the obvious shortest distance metric, i.e., the distance from $\sigma \in \mathbf{t}$ to $\sigma' \in \mathbf{t}$ is the shortest distance in $\mathbf{t}$, when following along the line segments of $\mathbf{t}$. Alternatively, we can identify (by loosing the order of $\mathbf{t}$) $\mathbf{t} = \mathcal{T}_{C(\mathbf{t})}$ where $C(\mathbf{t})$ is the contour function of $\mathbf{t}$ (see Figure 2.4) Via the inclusion $\mathbf{t} \in \mathbf{A} \hookrightarrow \mathcal{T}_{C(\mathbf{t})} \in \mathbb{T}$ one


 Figure 2.4: Tree $\mathbf{t}$ coded by its contour function $C(\mathbf{t})$.

2.3. Definition and properties of the continuum random tree

can view the space of finite rooted ordered trees $\mathbf{A}$ as a subspace of the space of all compact rooted $\mathbb{R}$ -trees $\mathbf{T}$.

Now I will give the construction of a *Galton-Watson tree*.

Construction 2.25 Let μ be a probability measure on the positive integers $\mathbb{N}$ such that

$$\sum_{k=0}^{\infty} k\mu(k) \leq 1,$$

and $\mu(1) \neq 1$. We will call such a μ a **(sub)critical offspring distribution**. Now let $\{K_u\}_{u \in \mathcal{U}}$ be a collection of independent $\mathbb{N}$ -valued random variables with law⁸ μ . A **Galton-Watson tree** θ **offspring distribution** μ is the random subset of $\mathcal{U}$ given by

$$\theta = \left\{ u = (u^1, \dots, u^n) \in \mathcal{U} \mid u^j \leq K_{(u^1, \dots, u^{j-1})} \text{ for every } 1 \leq j \leq n \right\}.$$

To justify that this in fact is a finite rooted tree, I will provide the proof of the following proposition. To prove this, I briefly recall what a *Galton-Watson process with offspring distribution* μ is:

Let $\{X_n^i\}_{(i,n) \in \mathbb{N} \times \mathbb{N}}$ be a collection of independent $\mathbb{N}$ -valued random variables with law μ . Now set $Z_0 = 1$ and $Z_1 = X_1^1$ and for $n \geq 0$

$$Z_{n+1} = \sum_{i=1}^{Z_n} X_n^i.$$

Then, the collection of random variables $\{Z_n\}_{n \geq 0}$ is called a **Galton-Watson process with offspring distribution** μ . The goal of the next two lemmas are to compute the probability-generating functions⁹ of Z_n and compute the extinction probability $P(Z_n = 0)$ as $n \rightarrow \infty$.

Lemma 2.26 Let $m_\mu(t) = \sum_{k \geq 0} t^k \mu(k)$ be the probability-generating function of law μ . Then, it holds that

$$m_{Z_n}(t) = \underbrace{(m_\mu \circ \dots \circ m_\mu)}_{n \text{ times}}(t), \quad n \geq 1. \quad (*)$$

Proof Since $Z_1 = X_1^1$ and X_1^1 has law μ $(*)$ holds for $n = 1$. Suppose now $(*)$ holds for some $n \geq 1$. Then we find by induction as X_n^i are indepent of

⁸A random variable X on a probability space $(\Omega, \mathcal{F}, P)$ has law $P \circ X^{-1}$.

⁹A random variable X has **probability-generating function** $m_X(t) = E[t^X]$.

Z_n and of each other:

$$\begin{aligned}
 m_{Z_{n+1}}(t) &= E \left[t^{Z_{n+1}} \right] \\
 &= \sum_{m \geq 0} E \left[t^{\sum_{i=0}^m X_n^i} \middle| Z_n = m \right] P(Z_n = m) \\
 &= \sum_{m \geq 0} E \left[t^{\sum_{i=0}^m X_n^i} \right] P(Z_n = m) \\
 &= \sum_{m \geq 0} m_\mu(t)^m P(Z_n = m) \\
 &= m_{Z_n}(m_\mu(t)) \\
 &= \underbrace{(m_\mu \circ \dots \circ m_\mu)}_{n+1 \text{ times}}(t).
 \end{aligned}
 \quad \square$$

The next lemma establishes a result about the extinction probability of a Galton-Watson process.

Lemma 2.27 *Let $p = \lim_{n \rightarrow \infty} P(Z_n = 0)$ and suppose μ is given as in Construction 2.25, then it holds that $p = 1$, i.e. the Galton-Watson process almost surely goes extinct.*

Proof By Lemma 2.26 we have that Z_n has probability-generating function

$$m_{Z_n}(t) = \underbrace{(m_\mu \circ \dots \circ m_\mu)}_{n \text{ times}}(t)$$

and we note that $m_{Z_n}(0) = P(Z_n = 0)$. We find that $p = \lim_{n \rightarrow \infty} m_{Z_n}(0)$. Note that if $Z_n = 0$, then also $Z_{n+1} = 0$. Therefore, it follows that $m_{Z_n}(0) \leq m_{Z_{n+1}}(0)$ for any $n \geq 1$. Therefore, p must be the smallest nonnegative fixed-point of the m_μ , i.e., the solution to the equation

$$m_\mu(t) = t.$$

Because, now $E[X_1^1] = \sum_{k=0}^{\infty} k\mu(k) \leq 1 = m'_\mu(1)$ it follows that $p = 1$. $\square$

Proposition 2.28 *θ as defined in Construction 2.25 is almost surely (a.s.) a finite rooted tree.*

Proof First, I show properties (t1)-(t3):

(t1) To show is that $\emptyset \in \theta$ a.s., i.e. $P(\emptyset \in \theta) = 1$. Because for $u = \emptyset$, the condition $u^j \leq K_{(u^1, \dots, u^{j-1})}$ is vacuous for any $j \in \mathbb{N}$ it follows that $P(\emptyset \in \theta) = 1$.

(t2) For $u \in \theta \setminus \{\emptyset\}$ we have

$$u^j \leq K_{(u^1, \dots, u^{j-1})} \text{ for every } 1 \leq j \leq n.$$

In particular, we have

$$u^j \leq K_{(u^1, \dots, u^{j-1})} \text{ for every } 1 \leq j \leq n-1,$$

which implies that $\pi(u) \in \theta$. That is, we have the following inclusions of events $\{u \in \theta \setminus \{\emptyset\}\} \subseteq \{\pi(u) \in \theta\}$ which shows that

$$P(u \in \theta \setminus \{\emptyset\} \implies \pi(u) \in \theta) = 1.$$

(t3) Consider the following events for $u \in \mathcal{U}$ and $n \in \mathbb{N}$

$$A_{u,j} = \{(u, j) \in \theta \iff 1 \leq j \leq K_u\}.$$

For $u \in \theta$ it follows that $A_{u,j} = \Omega$ (i.e. the whole probability space) and $P(A_{u,j}) = 1$. Then, by the independence of the K_u 's we find that:

$$\begin{aligned} P(\forall u \in \theta \forall j \in \mathbb{N} : (u, j) \in \theta \iff 1 \leq j \leq K_u) &= P\left(\bigcap_{(u,j) \in \mathcal{U} \times \mathbb{N}} A_{u,j}\right) \\ &= 1. \end{aligned}$$

This shows that for any $u \in \theta$ property (t3) holds and $\kappa_u(\theta) = K_u$.

For the finiteness of θ consider the following $\mathbb{N}$ -valued random variables

$$Z_n = \#\{u \in \theta \mid |u| = n\}, \quad n \geq 0.$$

Because only $\emptyset \in \theta$ has generation $|\emptyset| = 0$ it follows that $Z_0 = 1$. Moreover, we find that $Z_1 = K_\emptyset$. Now, we write for any $n \geq 0$

$$\{u \in \theta \mid |u| = n\} = \{u_n^i\}_{i=1}^{Z_n},$$

and note for any $n \geq 1$ that

$$\begin{aligned} Z_{n+1} &= \#\{u \in \theta \mid |u| = n+1\} \\ &\stackrel{(t3)}{=} \#\{(u', j) \in \theta \mid |u'| = n \text{ and } 1 \leq j \leq K_{u'}\} \\ &= \sum_{|u'|=n} K_{u'} \\ &= \sum_{i=1}^{Z_n} K_{u_n^i}. \end{aligned}$$

Because the random variables $K_{u_n^i}$ are independently distributed with law μ , it follows that $\{Z_n\}_{n \geq 0}$ is a Galton-Watson process with offspring distribution μ . Since μ is subcritical, we have by Lemma 2.27 that almost surely for n large enough $Z_n = 0$, which implies that θ is almost surely finite and hence a finite rooted tree. $\square$

Definition 2.29 A finite rooted ordered random tree $\theta \in \mathbf{A}$ as Construction 2.25 and Proposition 2.28 is called a **Galton-Watson process** with **offspring distribution** μ or simply μ -**Galton-Watson tree**. We denote by Π_μ the law of θ (viewed as a $\mathbf{A}$ -valued random variable, where $\mathbf{A} \subset \mathbb{T}$) on the space $\mathbf{A}$.

To make more sense of the law Π_μ of μ -Galton-Watson tree, I now give a explicit formula, where I follow [8].

Proposition 2.30 For any $\mathbf{t} \in \mathbf{A}$ it holds that

$$\Pi_\mu(\mathbf{t}) = \prod_{u \in \mathcal{T}} \mu(\kappa_u(\mathbf{t})).$$

Proof Let $\{K_u\}_{u \in \mathcal{U}}$ be the random variables that define a μ -Galton-Watson tree θ . Then, as shown in Proposition 2.28, we have that $\kappa_u(\theta) = K_u$ for all $u \in \theta$. Hence, it follows that

$$\{\theta = \mathbf{t}\} = \bigcap_{u \in \mathbf{t}} \{K_u = \kappa_u(\mathbf{t})\}.$$

Then, we can compute by independence of the K_u 's

$$\begin{aligned} \Pi_\mu(\mathbf{t}) &= P(\theta = \mathbf{t}) \\ &= \prod_{u \in \mathbf{t}} P(K_u = \kappa_u(\mathbf{t})) \\ &= \prod_{u \in \mathbf{t}} \mu(\kappa_u(\mathbf{t})). \end{aligned} \quad \square$$

I will introduce now the *contour process* of a sequence of Galton-Watson trees.

Construction 2.31 Let $\{\theta_n\}_{n \in \mathbb{N}}$ be a sequence of independent μ -Galton-Watson trees. Then the **contour process** $\{C_s\}_{s \geq 0}$ associated with $\{\theta_n\}_{n \in \mathbb{N}}$ by setting

$$C_s = C_{\bar{s}}(\theta_n), \text{ for } n \leq s < n+1,$$

where $\bar{s}$ denotes the fractional part of $s \geq 0$.

Next, I will state the result that the contour functions of a sequence of Galton-Watson trees *converges in distribution* to the a scaled Brownian excursion. I will not provide the proof for this result, as it is quite technical and will not give any better insight for the purpose of this thesis. I will though sketch the proof. For a detailed proof, I refer the reader to [8]. Recall that a sequence $\{X_n\}_{n \in \mathbb{N}}$ of (E, d) -random variables on a probability space $(\Omega, \mathcal{F}, P)$ **converges in distribution** to a (E, d) -random variable X , if for any $A \in \mathcal{B}(E)$ it holds that:

$$P(X_n \in A) \xrightarrow{n \rightarrow \infty} P(X \in A).$$

I will denote this type of convergence by $X_n \xrightarrow{(d)} X$ as $n \rightarrow \infty$.

2.3. Definition and properties of the continuum random tree

Theorem 2.32 For every $n \in \mathbb{N}$ let θ_n be the μ -Galton-Watson tree conditioned to have $\#(\theta_n) = n$ (That is its distribution is given by $\Pi_\mu(dt|\#(t) = n)$.) and let $\{e_s\}_{s \in [0,1]}$ be a Brownian excursion. Denote by $\{C_s\}_{s \geq 0}$ the contour process associated with $\{\theta_n\}_{n \in \mathbb{N}}$. Then, it holds that for any $s \in [0, 1]$

$$\frac{1}{\sqrt{2n}} C_{2ns} \xrightarrow[n \rightarrow \infty]{(d)} \frac{2}{\sigma} e_s,$$

where $\sigma^2 > 0$ is assumed to be the finite variance of a μ -Galton-Watson process.

Proof Sketch (i) First one can show that the contour process can be seen as a random walk $S = \{S_n\}_{n \in \mathbb{N}}$. This is clear from Figure 2.2 and the fact that the contour process is the concatenation of contour functions.

(ii) Then one can show that $\frac{1}{\sqrt{2n}} C_{2ns}$ converges to $\frac{2}{\sigma^2 \sqrt{2n}} S_{2ns}$ in probability as $n \rightarrow \infty$, where $\bar{\cdot}$ denotes the fractional part of a real number.

(iii) Then one shows that $\frac{2}{\sigma^2 \sqrt{2n}} S_{2ns}$ converges in distribution to $\frac{2}{\sigma} e_s$ as $n \rightarrow \infty$. $\square$

Equipped with Theorem 2.32 I can now state the following result regarding the convergence of Galton-Watson trees to the continuum random tree. For this, I fix the following notation: For a real tree $(\mathcal{T}, d_{\mathcal{T}}) \in \mathbb{T}$ and $\lambda > 0$ denote by $\lambda \mathcal{T}$ the real tree with the metric defined by $d_{\lambda \mathcal{T}} = \lambda d_{\mathcal{T}}$.

Theorem 2.33 For every $n \in \mathbb{N}$ let $\theta_n \in \mathbf{A} \subset \mathbb{T}$ be a μ -Galton-Watson tree conditioned to have $\#(\theta_n) = n$. Then $\frac{1}{\sqrt{2n}} \theta_n$ converges in distribution to the continuum random tree $\mathcal{T}_e$, in the metric space $\mathbb{T}$.

Proof Let C^n be the contour function associated with θ_n . Define $\hat{C}^n = \frac{1}{\sqrt{2n}} C^n$. Then, by the identification done in Remark 2.24 we have

$$\frac{1}{\sqrt{2n}} \theta_n = \mathcal{T}_{\hat{C}^n}.$$

By Theorem 2.32 we have that for any $s \in [0, 1]$ that

$$\hat{C}_s^n \xrightarrow[n \rightarrow \infty]{(d)} e_s.$$

Hence, since the tree map $\mathcal{T}_\bullet$ is continuous (by Theorem 2.11), it follows that $\frac{1}{\sqrt{2n}} \theta_n = \mathcal{T}_{\hat{C}^n}$ converges in distribution to the continuum random tree $\mathcal{T}_e$ in the metric space $\mathbb{T}$. $\square$

Remark 2.34 Theorem 2.33 is quite reminiscent of Donsker's theorem (see [5]), which asserts the existence of Brownian motion as the scaled limit of a simple random walk. Here, we asserted that CRT or Brownian tree $\mathcal{T}_e$ is the limit of a sequence of scaled Galton-Watson trees, which can be seen as the tree-analogue of a simple random walk.

This result concludes this long subsection about the CRT $\mathcal{T}_e$ as a distributional limit of Galton-Watson trees. This also concludes the definition of CRT and its convergence properties. In the next chapter, I will discuss the construction of the so-called *Brownian map* and its topological and geometric properties.

Chapter 3

The Brownian Map

This chapter will be devoted to the construction and exploration of the properties of a special metric space called *Brownian map*, which arises as a quotient of the continuum random tree $\mathcal{T}_e$. I will closely follow [2].

3.1 Construction of Brownian motion indexed by trees and Brownian snakes

In this section, I will detail the construction of *Brownian motion indexed by the CRT $\mathcal{T}_e$* via the notion of *Brownian snakes*. First, I will introduce *Brownian motion indexed by deterministic real trees*.

Definition 3.1 Let $(\mathcal{T}, d)$ be a rooted real tree with root $\rho \in \mathcal{T}$. A **Brownian motion indexed by $\mathcal{T}$** is defined as a Gaussian process $\mathcal{Z} = (\mathcal{Z}_\sigma)_{\sigma \in \mathcal{T}}$ on a probability space $(\Omega, \mathcal{F}, P)$ such that

- (i) $\mathcal{Z}_\rho = 0$.
- (ii) $E[\mathcal{Z}_\sigma] = 0$ for all $\sigma \in \mathcal{T}$.
- (iii) $E[(\mathcal{Z}_\sigma - \mathcal{Z}_{\sigma'})^2] = d(\sigma, \sigma')$ for all $\sigma, \sigma' \in \mathcal{T}$.

3.1.1 Deterministic snakes

I will now describe an equivalent view of Brownian motion indexed by trees following [9]. For this purpose, I will introduce a new process, which in essence can be viewed as Brownian motion indexed by a tree. The setting is as follows: Consider a function $g \in \mathcal{C}([0, 1])$, i.e. $g : [0, 1] \rightarrow \mathbb{R}$ is a nonnegative with $g(0) = g(1) = 0$. Additionally, assume that g is actually **Hölder continuous**, that is, there exists two constants $K, \gamma > 0$ such that for every $s, t \in [0, 1]$ it holds that

$$|g(s) - g(t)| \leq K |s - t|^\gamma.$$

For the next lemma, again consider the quantity

$$m_g(s, t) = \inf_{r \in [s \wedge t, s \vee t]} g(r)$$

for every $s, t \in [0, 1]$ introduced in Construction 2.7.

Lemma 3.2 *For every $n \in \mathbb{N}$, $s_1, \dots, s_n \in [0, 1]$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ it holds that*

$$\sum_{i,j=1}^n \lambda_i \lambda_j m_g(s_i, s_j) \geq 0.$$

Proof Fix $n \in \mathbb{N}$ and $s_1, \dots, s_n \in [0, 1]$ and let $t \geq 0$. For every $i, j \in \{1, \dots, n\}$ define the relation:

$$i \sim_t j \iff m_g(s_i, s_j) \geq t.$$

Note that for all $i, j, k \in I_t := \{i' \mid g(s_{i'}) \geq t\}$ we have:

- $i \sim_t i$ as $g(s_i) \geq t$ and so $m_g(s_i, s_i) = g(s_i) \geq t$. So $\sim_t$ is reflexive.
- Certainly $\sim_t$ is symmetric as $m_g(s_i, s_j) = m_g(s_j, s_i)$.
- If $i \sim_t j$ and $j \sim_t k$, i.e. $m_g(s_i, s_j) \geq t$ and $m_g(s_j, s_k) \geq t$, then we must have $m_g(s_i, s_k) \geq t$. Indeed: We have that

$$[s_i \wedge s_k, s_i \vee s_k] \subseteq [s_i \wedge s_j, s_i \vee s_j] \cup [s_j \wedge s_k, s_j \vee s_k]$$

and hence,

$$\begin{aligned} m_g(s_i, s_k) &= \inf_{r \in [s_i \wedge s_k, s_i \vee s_k]} g(r) \\ &\geq \inf_{r \in [s_i \wedge s_j, s_i \vee s_j] \cup [s_j \wedge s_k, s_j \vee s_k]} g(r) \\ &= \min \{m_g(s_i, s_j), m_g(s_j, s_k)\} \\ &\geq t. \end{aligned}$$

Therefore, $\sim_t$ is an equivalence relation on I_t . Now we compute:

$$\begin{aligned} \sum_{i,j=1}^n \lambda_i \lambda_j 1_{I_t} &= \sum_{i \sim_t j} \lambda_i \lambda_j \\ &= \sum_{i \in I_t} \left(\sum_{j \in [i]_{\sim_t}} \lambda_j \right)^2 \\ &\geq 0. \end{aligned}$$

Now, integrate this identity with respect to the Lebesgue measure dt on $[0, \infty)$:

$$\begin{aligned} \int_0^\infty \sum_{i,j=1}^n \lambda_i \lambda_j 1_{I_t} dt &= \sum_{i,j=1}^n \lambda_i \lambda_j \int_0^\infty 1_{\{t \leq m_g(s_i, s_j)\}} dt \\ &= \sum_{i,j=1}^n \lambda_i \lambda_j \int_0^{m_g(s_i, s_j)} dt \\ &= \sum_{i,j=1}^n \lambda_i \lambda_j m_g(s_i, s_j) \\ &\geq 0. \end{aligned}$$

This shows the result. $\square$

Now, we consider the following definition.

Definition 3.3 Let T be any set. We then call a function $m : T \times T \rightarrow \mathbb{R}$

- **symmetric** if $m(s, t) = m(t, s)$ for all $s, t \in T$.
- **non-negative definite** if for all $n \in \mathbb{N}$, $s_1, \dots, s_n \in T$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ it holds that

$$\sum_{i,j=1}^n \lambda_i \lambda_j m(s_i, s_j) \geq 0.$$

That is, in terms of Definition 3.3 for $T = [0, 1]$ the map

$$(s, t) \in T \times T \mapsto m_g(s, t) \in [0, 1]$$

is symmetric and non-negative definite. For symmetric non-negative functions, we obtain the following result (following [10]).

Proposition 3.4 Let T be a set and $m : T \times T \rightarrow \mathbb{R}$ a symmetric and nonnegative definite function. Then there exists a Gaussian process $\{X_t\}_{t \in T}$ with mean 0 and covariance function

$$E[X_s X_t] = m(s, t).$$

For this, we extend the notion of a Gaussian process: A **Gaussian process indexed by a set** T is a collection $\{X_t\}_{t \in T}$ of random variables such that for any finite subset $F \subseteq T$ and any subset $\{\lambda_t\}_{t \in F} \subseteq \mathbb{R}$ the random variable

$$\sum_{t \in F} \lambda_t X_t$$

is a Gaussian random variable.

For the proof, as for existence of Brownian motion, we use again Theorem 2.17.

Proof For any finite subset $F \subseteq T$ let m_F be the restriction of m to $F \times F$. We can think of m_F as the matrix $M_F = \{m(s, t)\}_{s, t \in F} \in \mathbb{R}^{F \times F}$. By assumption, this must be a symmetric and positive semidefinite matrix. Hence, we can consider P_F , the law of the multivariate normal distribution¹ with mean 0 and covariance matrix M_F on $\mathbb{R}^F$. Let now $G \subseteq F$ and consider the projection $\pi_G^F : \mathbb{R}^F \rightarrow \mathbb{R}^G$. Using that $M_F \circ (\pi_G^F)^{-1} = M_G$, we find that $P_F \circ (\pi_G^F)^{-1} = P_G$. Hence, the collection $\{P_F\}_{F \subseteq T \text{ finite}}$ is consistent. By Theorem 2.17, it follows that there exists a probability measure P on $(\mathbb{R}^T, \mathcal{B}(\mathbb{R}^T))$ such that $P_F = P \circ (\pi_F^T)^{-1}$ for all finite subsets $F \subseteq T$. Now consider the coordinate process $X_t(\omega) = \omega(t)$ for any $\omega \in \mathbb{R}^T$ and $t \in T$. By construction, we have $X_t \sim \mathcal{N}(0, m(t, t))$. Note that for any $s, t \in T$ we have $(X_s, X_t) = \pi_{\{s, t\}}^T$ and so again by construction $(X_t, X_s) \sim \mathcal{N}(0, M_{\{s, t\}})$ and so it follows immediately that $E[X_s X_t] = m(s, t)$. That is $\{X_t\}_{t \in T}$ is a Gaussian process. $\square$

To prove the next lemma, we need a result establishing the continuity of the stochastic process. This result is due to Kolmogorov and can be found in [5].

Theorem 3.5 *Let $X = \{X_t\}_{t \geq 0}$ be $\mathbb{R}$ -valued stochastic process for which there exist three constants $\alpha, \beta, C > 0$ such that*

$$E[|X_s - X_t|^\alpha] \leq C |s - t|^\beta.$$

Then, there exists a continuous stochastic process $\tilde{X} = \{\tilde{X}_t\}_{t \geq 0}$ such that $X_t = \tilde{X}_t$ almost surely for all $t \geq 0$.

Using now Lemma 3.2 we find the following corollary.

Corollary 3.6 *There exists a continuous Gaussian process $Z^g = \{Z_s^g\}_{s \in [0, 1]}$ with mean 0 and covariance function*

$$E[Z_s^g Z_t^g] = m_g(s, t).$$

Moreover, $E[(Z_s^g - Z_t^g)^2] = d_g(s, t)$.

Proof By Lemma 3.2 the function $m_g : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ is symmetric and non-negative definite. Therefore, by Proposition 3.4 applied to $T = [0, 1]$ and $m = m_g$ we find such a Gaussian process Z with mean 0 and a covariance

¹A $\mathbb{R}^n$ -valued random variable X is multivariate normally distributed with mean $\mu \in \mathbb{R}^n$ and positive semidefinite covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$ if

$$P(X \in A) = \int_A \varphi_{\mu, \Sigma}(x) dx, A \in \mathcal{B}(\mathbb{R}^n),$$

where

$$\varphi_{\mu, \Sigma}(x) = (2\pi)^{-\frac{n}{2}} \exp\left(-\frac{1}{2} \langle (x - \mu), \Sigma^{-1}(x - \mu) \rangle\right).$$

We write $X \sim \mathcal{N}(\mu, \Sigma)$.

3.1. Construction of Brownian motion indexed by trees and Brownian snakes

function $E[Z_s Z_t] = m_g(s, t)$. Now note that we have the following bound by the Hölder continuity of g for $s, t \in [0, 1]$ and $r \in [s \wedge t, s \vee t]$ such that $g(r) = m_g(s, t)$:

$$\begin{aligned} E[(Z_s - Z_t)^2] &= E[Z_s^2] + E[Z_t^2] - 2E[Z_s Z_t] \\ &= m_g(s, s) + m_g(t, t) - 2m_g(s, t) \\ &= g(s) + g(t) - 2m_g(s, t) (= d_g(s, t)) \\ &= (g(s) - g(r)) + (g(t) - g(r)) \\ &\leq K |s - r|^\gamma + K |r - t|^\gamma \\ &\stackrel{r \in [s \wedge t, s \vee t]}{\leq} 2K |s - t|^\gamma. \end{aligned}$$

By Theorem 3.5 for $\alpha = 2, \beta = \max \gamma - 1, 0, C = 2K$, there exists a continuous stochastic process $Z^g = \{Z_s^g\}_{s \in [0, 1]}$ such that $Z_t^g = Z_s$ almost surely. Then Z^g is a continuous Gaussian process with mean 0 and covariance function $E[Z_s^g Z_t^g] = m_g(s, t)$. $\square$

Remark 3.7 As indicated in the last computation of the proof of Corollary 3.6 we have, in particular, that

$$E[(Z_s^g - Z_t^g)^2] = d_g(s, t).$$

where d_g is the metric of the real tree $\mathcal{T}_g$. Also note that since $E[(Z_s^g)^2] = g(s)$ it follows that $Z_0^g = Z_1^g = 0$ as $g(0) = g(1) = 0$.

Definition 3.8 The *snake driven by a Hölder continuous function* $g \in \mathcal{C}([0, 1])$ is the centered² Gaussian process $Z^g = \{Z_s^g\}_{s \in [0, 1]}$ and covariance function

$$E[Z_s^g Z_t^g] = m_g(s, t), \quad s, t \in [0, 1].$$

Remark 3.9 A classical result for centered Gaussian processes states that any centered Gaussian process X with X_0 is uniquely determined by its covariance function (see e.g. [5]). Therefore, the snake driven by g is uniquely determined by the function g .

The next lemma establishes the equivalence of Brownian motion indexed by the tree $\mathcal{T}_g$ with the snake driven by g .

Lemma 3.10 For any $s, t \in [0, 1]$ it holds that:

$$s \sim_g t \iff Z_s^g = Z_t^g \text{ almost surely.}$$

Proof For all $s, t \in [0, 1]$ we have the following equivalence:

$$\begin{aligned} s \sim_g t &\iff d_g(s, t) = 0 \\ &\iff E[(Z_s^g - Z_t^g)^2] = 0 \end{aligned}$$

²i.e. with mean 0.

Then, clearly for any $\varepsilon > 0$ we find by Chebyshev's inequality³:

$$\begin{aligned} P(|Z_s^g - Z_t^g| \geq \varepsilon) &\leq \frac{E[(Z_s^g - Z_t^g)^2]}{\varepsilon^2} \\ &= 0. \end{aligned}$$

Hence, $P(|Z_s^g - Z_t^g| \geq \varepsilon) = 0$ and therefore $P(|Z_s^g - Z_t^g| < \varepsilon) = 1$ for any $\varepsilon > 0$. From this, it follows that $Z_s^g = Z_t^g$ almost surely. $\square$

Remark 3.11 Let $g \in \mathcal{C}([0, 1])$ be a Hölder continuous and Z^g the snake driven by g . By Lemma ??, for any $s \in [0, 1]$ the value of Z_s^g only depends on the equivalence class of s with respect to the equivalence class $\sim_g$, which defines the real tree $\mathcal{T}_g$. Hence, we can think of Z^g as a Brownian motion indexed by $\mathcal{T}_g$ the following way: Let $p_g : [0, 1] \rightarrow \mathcal{T}_g$ be the canonical projection. Then, we can define for any $s \in [0, 1]$ and $\sigma = p_g(s)$ the random variable $\mathcal{Z}_\sigma := Z_s^g$. This is well-defined by Lemma 3.10. Indeed, we have for any $s, t \in [0, 1]$ and $\sigma = p_g(s)$ and $\sigma' = p_g(t)$:

$$\begin{aligned} \sigma = \sigma' &\iff s \sim_g t \\ &\iff Z_s^g = Z_t^g \\ &\iff \mathcal{Z}_\sigma = \mathcal{Z}_{\sigma'}. \end{aligned}$$

Since Z^g is centered Gaussian, also $\mathcal{Z} = \{\mathcal{Z}_\sigma\}_{\sigma \in \mathcal{T}_g}$ is Gaussian with mean 0 and as $Z_0^g = 0$ we have for the root $\rho = p_g(0)$ that $\mathcal{Z}_\rho = 0$. By Remark 3.7, we have that for all $\sigma = p_g(s), \sigma' = p_g(t) \in \mathcal{T}_g$ that

$$\begin{aligned} E[(\mathcal{Z}_\sigma - \mathcal{Z}_{\sigma'})^2] &= E[(Z_s^g - Z_t^g)^2] \\ &= d_g(s, t) \\ &= d_g(\sigma, \sigma'). \end{aligned}$$

This shows that $\mathcal{Z}$ is a Brownian motion indexed by $\mathcal{T}_g$.

Conversely, if $\mathcal{Z}$ is a Brownian motion indexed by $\mathcal{T}_g$. Define for all $s \in [0, 1]$ the random variable $Z_s := \mathcal{Z}_{p_g(s)}$. By definition of Brownian motion indexed by $\mathcal{T}_g$, the process $Z = \{Z_s\}_{s \in [0, 1]}$ is a centered Gaussian process. Since $p_g(0) = \rho$ we have $Z_0 = \mathcal{Z}_\rho = 0$. Hence, we can compute

$$\begin{aligned} E[Z_s^2] &= E[(Z_s - Z_0)^2] \\ &= E[(\mathcal{Z}_{p_g(s)} - \mathcal{Z}_{p_g(0)})^2] \\ &= d_g(p_g(s), p_g(0)) \\ &= g(s). \end{aligned}$$

From this, we can compute the covariance function of Z for $s, t \in [0, 1]$:

$$\begin{aligned} E[(Z_s - Z_t)^2] &= d_g(p_g(s), p_g(t)) \iff -2E[Z_s Z_t] = -2m_g(s, t) \\ &\iff E[Z_s Z_t] = m_g(s, t). \end{aligned}$$

³ $P(|X - E[X]| \geq \varepsilon) \leq \frac{E[(X - E[X])^2]}{\varepsilon^2}$.

This shows that Z is the snake driven by g .

It turns out to be quite practical, to view a Brownian motion indexed by $\mathcal{T}_g$ as the snake driven by g , especially if g becomes a random function, e.g. Brownian excursion.

This concludes this short introduction to snakes driven by deterministic functions.

3.1.2 Brownian snake and Brownian motion indexed by the CRT

In this subsection, I introduce the *Brownian snake* and its connection to Brownian motion indexed by CRT.

Remark 3.12 *From classical results on the continuity of paths of Brownian motion, we know already that Brownian motion has Hölder continuous paths. Hence, a Brownian excursion $e = \{e_s\}_{s \in [0,1]}$*

By Remark 3.12 the following definition is justified.

Definition 3.13 *Let $e = \{e_s\}_{s \in [0,1]}$ be a Brownian excursion, where we view e as the $\mathcal{C}([0,1])$ -valued random variable. Then, conditionally on e , we can consider the stochastic process $Z^e = \{Z_s^e\}_{s \in [0,1]}$, i.e., for all $s \in [0,1]$*

$$Z_s^e = E[Z_s^e | \sigma(e)],$$

*where $\sigma(e)$ is the σ -algebra generated by e . We call the stochastic process Z^e the **Brownian snake (driven by e)**.*

Remark 3.14 *To make the connection note that conditionally on e , the process Z^e is a centered Gaussian process with (conditional) covariance function*

$$E[Z_s^e Z_t^e | \sigma(e)] = m_e(s, t).$$

By Remark 3.11 Z^e conditional on e can be viewed as Brownian motion $Z = \{Z_\sigma\}_{\sigma \in \mathcal{T}_e}$ indexed by the continuum random tree $\mathcal{T}_e$. In this way, we avoid the technical difficulties in considering a stochastic process indexed by the random set $\mathcal{T}_e$. From now on, when talking about Brownian motion indexed by CRT, I identify it with the Brownian snake driven by e .

3.2 Brownian map and the homeomorphism result

The preliminary results of the preceding section enable us now to construct the object of interest of this chapter, the so-called Brownian map. I will follow the construction given in [11]. But we begin by a reidentification of the terms we have introduced so far and discuss the quotients of real trees in a new light.

3.2.1 Quotients of real trees homeomorphic to the 2-sphere $\mathbb{S}^2$

We identify the unit interval $[0, 1]$ by the unit circle in the complex plane $\mathbb{C}$

$$\mathbb{S}^1 = \{a \in \mathbb{C} \mid |a| = 1\} \subseteq \mathbb{C}$$

by the map

$$e^{2i\pi t} \in \mathbb{S}^1 \mapsto t \in [0, 1].$$

For $a, b \in \mathbb{C}$ we set $[a, b]$ to be the **closed arc** in $\mathbb{S}^1$ going from a to b in clockwise direction with the convention $[a, a] = \{a\}$. $(a, b) := [a, b] \setminus \{a, b\}$ will be called the **open arc** with the convention $(a, a) = \emptyset$. Under this identification we can reidentify the following notions:

- (i) A continuous non-negative function $g : [0, 1] \rightarrow \mathbb{R}$ with $g(0) = g(1) = 0$ can be viewed as a continuous non-negative function $g : \mathbb{S}^1 \rightarrow \mathbb{R}$ with $g(1) = 0$. Denote by $\mathcal{C}(\mathbb{S}^1)$ the space of such function with the supremum metric.
- (ii) We identify the quantity $m_g(s, t)$ for $s, t \in [0, 1]$ and maps $g \in \mathcal{C}([0, 1])$ by the following equivalent quantity for $\mathcal{C}(\mathbb{S}^1)$

$$m_g(a, b) = \max \left\{ \min_{c \in [a, b]} g(c), \min_{c \in [b, a]} g(c) \right\}, \quad a, b \in \mathbb{S}^1.$$

- (iii) Similarly for $g \in \mathcal{C}(\mathbb{S}^1)$ we define the pseudometric

$$d_g(a, b) = g(a) + g(b) - 2m_g(a, b), \quad a, b \in \mathbb{S}^1,$$

and the equivalence relation $\sim_g$ for $a, b \in \mathbb{S}^1$ by

$$a \sim_g b \iff d_g(a, b) = 0.$$

- (iv) This way we can identify the tree $\mathcal{T}_g$ coded by $g \in \mathcal{C}([0, 1])$ as the quotient $\mathcal{T}_g = \mathbb{S}^1 / \sim_g$ coded by $g \in \mathcal{C}(\mathbb{S}^1)$, which has the root $\rho = p_g(1)$, where $p_g : \mathbb{S}^1 \rightarrow \mathcal{T}_g$ is the canonical projection.

(Hyperbolic) geodesic laminations

We denote by $\mathbb{D} \subseteq \mathbb{C}$ the open unit disk in the complex plane and $\overline{\mathbb{D}} = \mathbb{D} \cup \mathbb{S}^1$ the closed unit disk. We equip $\mathbb{D}$ with the usual **hyperbolic metric**, that is for $a, b \in \mathbb{D}$ their distance is given by

$$d_{\mathbb{D}}(a, b) := \inf_{\gamma} L(\gamma),$$

where

- the infimum is taken over piecewise differentiable paths $\gamma : [0, 1] \rightarrow \mathbb{D}$ with $\gamma(0) = a$ and $\gamma(1) = b$ and
- for such path γ denote by

$$L(\gamma) := \int_0^1 \frac{2|\gamma'(t)|}{1 - |\gamma(t)|^2} dt$$

its (hyperbolic) path length.

We say that path $\gamma : [0, 1] \rightarrow \mathbb{D}$ is a **(hyperbolic minimizing) geodesic** if

$$\forall t_1, t_2 \in [0, 1] \cap [0, 1] : d_{\mathbb{D}}(\gamma(t_1), \gamma(t_2)) = |t_1 - t_2|.$$

Such a geodesic is unique. Denote the **open** resp. **closed** geodesic lines by

$$\gamma(0)\gamma(1) := \gamma([0, 1] \setminus \{0, 1\})$$

$$\overline{\gamma(0)\gamma(1)} = \gamma([0, 1]).$$

Now we will introduce the concept of a geodesic lamination. This is following closely [12].

Definition 3.15 *A (hyperbolic) geodesic lamination in $\mathbb{D}$ is a non-empty closed subset $\mathcal{L} \subset \mathbb{D}$, which is the disjoint union of a family of geodesic lines, i.e.*

$$\mathcal{L} = \bigsqcup_{i \in I} \gamma_i([0, 1]),$$

where $\{\gamma_i : [0, 1] \rightarrow \mathbb{D}\}_{i \in I}$ are geodesics such that $\gamma_i([0, 1]) \cap \gamma_j([0, 1]) = \emptyset$ for all $i \neq j \in I$. We say that a lamination $\mathcal{L}$ in $\mathbb{D}$ is maximal if for any other lamination $\mathcal{L}'$ it holds that $\mathcal{L}' \subset \mathcal{L}$.

In the setting of the equivalence relation $\sim_g$ for $g \in \mathcal{C}(S^1)$ on the unit circle $S^1 \subset \mathbb{C}$, we define the following subset of $\mathbb{D}$

$$\mathcal{L}_g := \bigcup_{a, b \in S^1 : a \neq b, a \sim_g b} ab \subseteq \mathbb{D}.$$

Furthermore, we introduce the following additional assumption on the function $g : S^1 \rightarrow \mathbb{R}$:

Assumption 3.16 *We say g satisfies the assumption (H_g) if local minima of g are distinct, i.e., for any two disjoint open arcs $(a, b), (c, d) \subset S^1$ such that g attains its infimum on (a, b) resp. (c, d) at a point in (a, b) resp. (c, d) , then*

$$\min_{x \in [a, b]} g(x) \neq \min_{x \in [c, d]} g(x).$$

Now, under (H_g) we have the following property of the equivalence relation $\sim_g$.

Lemma 3.17 *Under the assumption (H_g) any equivalence class for $\sim_g$ can only contain at most three points.*

Proof Suppose in contradiction, that there is an equivalence class $\mathcal{E}$ for $\sim_g$ containing more than three points. Let $a, b, c, d \in \mathcal{E}$ be four distinct points in this equivalence class. Then note that we have for any $x \neq y$ in $\{a, b, c, d\}$ that

$$g(x) = g(y) = m_g(x, y) = \max \left\{ \min_{z \in [x, y]} g(z), \min_{z \in [y, x]} g(z) \right\}.$$

In particular, we have $g(a) = g(b) = g(c) = g(d)$. If now (a, b) and (c, d) are not disjoint, suppose without loss of generality that the arc (a, b) comes before the arc (c, d) in counterclockwise direction and overlaps at c and b (see Figure 3.1 below), then clearly we have that (c, d) and (d, c) are disjoint open arcs (in blue resp. green in Figure 3.1), where g attains its minimum at $g(b)$ for $b \in (c, d)$ and $g(a)$ for $a \in (d, c)$. As $g(a) = g(b)$, this constitutes a contradiction. If now (a, b) and (c, d) are disjoint open arcs (as in Figure

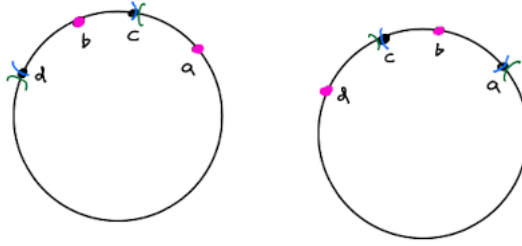


Figure 3.1: Non-disjoint arcs (a, b) and (c, d) . **Figure 3.2:** Disjoint arcs (a, b) and (c, d) .

3.2), then we clearly have that (a, c) and (c, a) disjoint open arcs (in blue resp. green in Figure 3.2), where g attains its minimum at $g(b)$ for $b \in (a, c)$ and at $g(d)$ for $d \in (c, a)$. This is again a contradiction. Therefore, an equivalence class for $\sim_g$ has at most three points. $\square$

Remark 3.18 *It should be noted that Lemma 3.17 also shows that any vertex σ of a real tree $\mathcal{T}_g$ coded by $g \in \mathcal{C}(\mathbb{S}^1)$ satisfying (H_g) satisfies that the cardinality of the preimage $p_g^{-1}(\{\sigma\})$ is at most three.*

The next proposition will establish that $\mathcal{L}_g$ under assumption (H_g) on g defines a geodesic lamination. The proof follows [11].

Proposition 3.19 *Under the assumption (H_g) , the set $\mathcal{L}_g$ is a maximal geodesic lamination in $\mathbb{D}$.*

Proof Let $\{a, b\}, \{c, d\}$ be two distinct pairs of distinct points in $\mathbb{S}^1$ such that $a \sim_g b$ and $c \sim_g d$. We have the following claim:

Claim *Either (a, b) and (c, d) are disjoint arcs or one contains the other.*

Proof Suppose this is not the case, then we are in the case depicted in Figure 3.1. Since $g(b) = m_g(a, b)$ and $g(c) = m_g(c, d)$, we must also have $g(b) = g(c) = m_g(b, c)$ and hence $b \sim_g c$. From this follows by transitivity of $\sim_g$, that

$$a \sim_g b \sim_g c \sim_g d.$$

That is a, b, c, d are four distinct points in the same equivalence class for $\sim_g$. By Lemma 3.17 this is a contradiction under the assumption (H_g) . $\square$

By the claim, we can conclude that the geodesic lines ab and cd are disjoint for distinct pairs of distinct points and coincide if $\{a, b\} = \{c, d\}$. Hence, the set $\mathcal{L}_g$ is the disjoint union of geodesic lines. Next, we claim that $\mathcal{L}_g$ is closed.

Claim $\mathcal{L}_g$ is a closed subset of $\mathbb{D}$.

Proof The graph $\{(x, y) \in \mathbb{S}^1 \times \mathbb{S}^1 \mid x \sim_g y\}$ of $\sim_g$ is closed as for any (x, y) in its complements, we can pick a small enough neighbourhood U around x and V around y in $\mathbb{S}^1$ such that for any $(x', y') \in U \times V$ we have $x' \not\sim_g y'$. Hence, the complement of the graph is closed. This shows that $\mathcal{L}_g$ is a closed subset. $\square$

This shows that $\mathcal{L}_g$ is a geodesic lamination in $\mathbb{D}$. Now, we will show that it is maximal among all geodesic laminations:

Suppose in contradiction that $a, b \in \mathbb{S}^1$ are distinct points such that the geodesic line $ab \subset \mathbb{D}$ does not intersect $\mathcal{L}_g$. Without loss of generality, assume that

$$\min_{x \in [a, b]} g(x) \geq \min_{x \in [b, a]} g(x).$$

Assume now $g(a) > \min_{x \in [a, b]} g(x)$. For $\varepsilon > 0$ small enough we find $c = e^{i\varepsilon}a \in (a, b)$ and $d = e^{-i\varepsilon}a \in (b, a)$ such that

$$g(a) > g(c) = g(d) = \min_{x \in [c, d]} g(x) > \min_{x \in [a, b]} g(x).$$

For this see the Figure 3.3 below. However, this implies that $c \sim_g d$ and therefore the geodesic lines $cd \subset \mathcal{L}_g$ and ab intersect, which contradicts the assumption, that ab does not intersect $\mathcal{L}_g$. Therefore, we conclude

$$g(a) = \min_{x \in [a, b]} g(x).$$

By the same argument, we find that

$$g(b) = \min_{x \in [a, b]} g(x).$$

Hence, it follows that $a \sim_g b$, that is $ab \in \mathcal{L}_g$, which again is a contradiction. Therefore, we must have that any geodesic line ab intersects $\mathcal{L}_g$. This shows that $\mathcal{L}_g$ in fact is maximal among all the geodesic laminations of $\mathbb{D}$.



Figure 3.3: Small ε -arc.

The next corollary characterizes the components of the complement of the lamination $\mathcal{L}_g$

Corollary 3.20 *Each connected component of $\mathbb{D} \setminus \mathcal{L}_g$ under the assumption (H_g) is an ideal hyperbolic triangle.*

Remark 3.21 *An ideal hyperbolic triangle in $\mathbb{D}$ is a triangle, whose vertices lie all on the boundary S^1 (i.e. points at infinity).*

Proof Let T be a connected component of $\mathbb{D} \setminus \mathcal{L}_g$. Then, we must have that its boundary ∂T is a subset of $\mathcal{L}_g$. Hence it must be the disjoint union of geodesic lines in $\mathcal{L}_g$, which have endpoints on S^1 . By the Lemma 3.17 there are at most three such points under (H_g) . So we find as seen in Figure 3.4, two cases since the boundary must bound a connected component of $\mathbb{D} \setminus \mathcal{L}_g$. In either case, we have an ideal hyperbolic triangle T with boundary $\partial T = ab \cup bc \cup ca$ or $\partial T = ab \cup a\infty \cup b\infty$, where ∞ is the point at infinity of $\mathbb{D}$.

Remark 3.22 *By the Corollary 3.20 any component of $\mathbb{D} \setminus \mathcal{L}_g$ is in correspondence with a triple $\{a, b, c\}$ of distinct points in S^1 such that that $a \sim_g b \sim_g c$.*

The lamination $\mathcal{L}_g$ allows us to extend the equivalence relation $\sim_g$ on S^1 to $\overline{\mathbb{D}} = \mathbb{D} \cup S^1$.

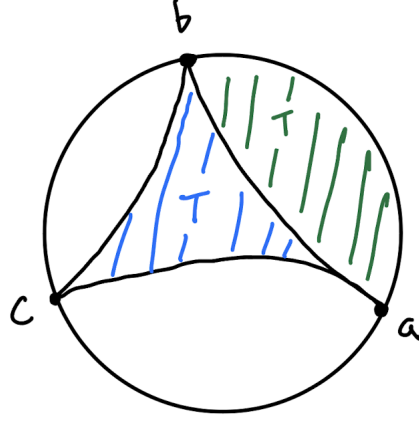


Figure 3.4: Complement of lamination.

Construction 3.23 The extension of $\sim_g$ to $\overline{\mathbb{D}}$ goes as follows: For $x, y \in \overline{\mathbb{D}}$ we define

$$x \sim_g y \iff \begin{cases} x, y \in \overline{ab}, \text{ for some arc } \overline{ab} \text{ with } a, b \in \mathbb{S}^1, a \sim_g b \\ x, y \in \text{cl } T, \text{ for some component } T \text{ of } \mathbb{D} \setminus \mathcal{L}_g. \end{cases}$$

This defines an equivalence relation on $\overline{\mathbb{D}}$, that extends the equivalence relation $\sim_g$ on $\mathbb{S}^1$. Indeed, reflexivity, symmetry, and transitivity can be easily checked. And note that if $x, y \in \mathbb{S}^1$ with $x \sim_g y$ (in the original equivalence relation on $\mathbb{S}^1$) we clearly have $x, y \in \overline{xy}$, so that $x \sim_g y$ also for the extended relation. Note that there are three possible types of equivalence classes for the extended equivalence relation $\sim_g$:

- (i) Singletons $\{a\}$ for $a \in \mathbb{S}^1$,
- (ii) closed arcs $\overline{ab}$ for $a, b \in \mathbb{S}^1$ with $a \neq b$ and $a \sim_g b$, or
- (iii) closures of ideal hyperbolic triangles with ends $a, b, c \in \mathbb{S}^1$ with $a \sim_g b \sim_g c$.

Note that we can identify the tree $\mathcal{T}_g$ as quotient of $\overline{\mathbb{D}}$.

Lemma 3.24 There is a homeomorphism $\mathcal{T}_g \rightarrow \overline{\mathbb{D}}/\sim_g$.

Proof Consider the inclusion $f : \mathbb{S}^1 \rightarrow \overline{\mathbb{D}}$. Then, clearly we have bijection

$$\tilde{f} : \mathcal{T}_g \rightarrow \overline{\mathbb{D}}/\sim_g$$

given by simply sending the equivalence class of any $a \in \mathbb{S}^1$ under $\sim_g$ on $\mathbb{S}^1$ to the equivalence class of $f(a) = a$ under $\sim_g$ on $\overline{\mathbb{D}}$. This is clearly a surjection by Construction 3.23 and the mentioned equivalence classes of

the extended relation. Obviously, if $a \sim_g b$ in $\overline{\mathbb{D}}$, then as $\sim_g$ is the extension of $\sim_g$ on $\mathbb{S}^1$, it follows that $a \sim_g b$ on $\mathbb{S}^1$. So $\tilde{f}$ is injective. Since $\tilde{f}$ is continuous and $\mathcal{T}_g$ and $\overline{\mathbb{D}}/\sim_g$ are both Hausdorff and compact, it follows that $\tilde{f}$ is a homeomorphism. $\square$

We now introduce another assumption.

Assumption 3.25 For $g, h \in \mathcal{C}(\mathbb{S}^1)$ we say assumption $(H'_{g,h})$ holds if for all $a, b, c \in \mathbb{S}^1$ with $a \sim_g b$ and $a \sim_h c$ it holds that

$$a = b \text{ or } a = c$$

Last but not least, in preparation for the main result of this subsection, for $g, h \in \mathcal{C}(\mathbb{S}^1)$ satisfying (H_g) , (H_h) and $(H'_{g,h})$ we declare the following equivalence relation $\approx_h$ on $\mathcal{T}_g$ by:

For $\alpha, \beta \in \mathcal{T}_g$ we set $\alpha \approx_h \beta$ if and only if there are representatives $a, b \in \mathbb{S}^1$ of α resp. β such that $a \sim_h b$. We check that this is in fact an equivalence relation.

Lemma 3.26 Under (H_g) , (H_h) and $(H'_{g,h})$ the relation $\approx_h$ is an equivalence relation on $\mathcal{T}_g$.

Proof Let $\alpha, \beta, \gamma \in \mathcal{T}_g$. Clearly, we have for any representative $a \in \mathbb{S}^1$ of α , that $a \sim_h a$ by reflexivity of $\sim_h$. Thus, $\alpha \approx_h \alpha$. This shows the reflexivity. Now, by definition $\approx_h$ is symmetric. Now if $\alpha \approx_h \beta$ and $\beta \approx_h \gamma$, there exist representative $a, c \in \mathbb{S}^1$ of α resp. γ and representatives $b, b' \in \mathbb{S}^1$ of β such that $a \sim_h b$ and $b' \sim_h c$. Since both b and b' are representatives of $\beta \in \mathcal{T}_g$ we must have that $b \sim_g b'$. By the assumption $(H'_{g,h})$ it must follow that:

- Since $a \sim_h b$ and $b \sim_g b'$ we must have $b = a$ or $b = b'$.
- Since $b' \sim_h c$ and $b \sim_g b'$ we must have $b' = c$ or $b = b'$.

If $b = b'$, then we have $a \sim_h c$ by the transitivity of $\sim_h$, and therefore $\alpha \approx_h \gamma$. If $b \neq b'$, then we have $a = b$ and $b' = c$, so we must have $\alpha = \beta = \gamma$ and therefore $\alpha \approx_h \gamma$. This shows transitivity. $\square$

Now comes the main result about the quotients of deterministic trees, the proof of which will follow the one given in [11] very closely.

Proposition 3.27 Under the assumptions (H_g) , (H_h) and $(H'_{g,h})$ the quotient space $\mathcal{T}_g/\approx_h$ is homeomorphic to the sphere $\mathbb{S}^2$.

Proof First, we embed the complex plane $\mathbb{C}$ in $\mathbb{R}^3$ by identifying it with the plane

$$\{x_3 = 0\} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 | x_3 = 0\}.$$

Define the closed upper hemisphere by

$$H_+ = \{(x_1, x_2, x_3) \in \mathbb{S}^2 \mid x_3 \geq 0\},$$

and the closed lower hemisphere by

$$H_- = \{(x_1, x_2, x_3) \in \mathbb{S}^2 \mid x_3 \leq 0\}.$$

By the stereographic projection from the north and south poles we can identify the closed unit disk $\overline{\mathbb{D}}$ homeomorphically with H_+ and H_- . By Lemma 3.24 we have that $H_+/\sim_g$ is homeomorphic to $\mathcal{T}_g$ and $H_-/\sim_h$ is homeomorphic to $\mathcal{T}_h$. Let now $\sim$ be the relation on $\mathbb{S}^2$ defined by

$$x \sim y \iff x \sim_g y \text{ or } x \sim_h y, \quad x, y \in \mathbb{S}^2.$$

This relation is clearly reflexive and symmetric. Now, if for $x, y, z \in \mathbb{S}^2$ we have $x \sim y$ and $y \sim z$. Then, we have the following possibilities:

- If both $x \sim_g y$ and $y \sim_g z$ or $x \sim_h y$ and $y \sim_h z$ we are done. Indeed, by the transitivity of g or h , we have $x \sim z$.
- If $x \sim_g y$ and $y \sim_h z$, then by the assumption $(H'_{g,h})$ we have $y = x$ or $y = z$. In both cases, we find $x \sim z$.
- The case where $x \sim_h y$ and $y \sim_g z$ works analogously.

So $\sim$ is an equivalence relation on $\mathbb{S}^2$. Because both the relations $\sim_g$ and $\sim_h$ are closed, i.e., their graph is closed⁴, also the equivalence relation $\sim$ is closed. Moreover, in Construction 3.23 we have identified the equivalence classes of $\sim_g$ and $\sim_h$ as singletons, arcs, or closures of an ideal hyperbolic triangles. Since an equivalence class for $\sim$ is an equivalence class for $\sim_g$ or for $\sim_h$, it follows that any equivalence class for $\sim$ is a compact path-connected subset of $\mathbb{S}^2$, whose complement is also path-connected (i.e. $\mathbb{S}^2$ minus a point, minus a arc, or minus the closure of an ideal hyperbolic triangle.). Now, we will use the following result by Moore [13]:

Theorem 3.28 *Let $\sim$ be a closed equivalence relation on $\mathbb{S}^2$. If every equivalence class of $\sim$ is a compact path-connected subset of $\mathbb{S}^2$ with path-connected complement, then the quotient space $\mathbb{S}^2/\sim$ is homeomorphic to $\mathbb{S}^2$.*

Based on the above arguments, our equivalence relation $\sim$ on $\mathbb{S}^2$ satisfies the premise of this theorem. Hence, $\mathbb{S}^2/\sim$ is homeomorphic to $\mathbb{S}^2$. First note that by composing the canonical projections

$$\mathbb{S}^1 \rightarrow \mathcal{T}_g \rightarrow \mathcal{T}_g/\approx_h$$

⁴A equivalence relation $\sim$ on a topological space X is closed if its graph $\{(x, y) \in X \times X \mid x \sim y\}$ is a closed subset of $X \times X$.

we find that $\mathcal{T}_g/\approx_h$ is compact. The composition of inclusion and canonical projection

$$\mathbb{S}^1 \rightarrow \mathbb{S}^2 \rightarrow \mathbb{S}^2/\sim$$

then factors through the equivalence relation $\sim_g$ to a continuous surjection $\mathcal{T}_g \rightarrow \mathbb{S}^2/\sim$, which in turn again factors through the equivalence relation $\approx_h$ to a continuous bijection $\mathcal{T}_g/\approx_h \rightarrow \mathbb{S}^2/\sim$. By compactness of $\mathcal{T}_g/\approx_h$ and $\mathbb{S}^2/\sim$ it follows that this bijection must be a homeomorphism. Therefore, from the result derived from Theorem 3.28 it follows that $\mathcal{T}_g/\approx_h$ is homeomorphic to $\mathbb{S}^2$. $\square$

This concludes the homeomorphism result for deterministic real trees. We now turn to the construction of the Brownian map.

3.2.2 Construction of the Brownian map

Recall from Definition 3.13 that for a Brownian excursion $e = \{e_s\}_{s \in [0,1]}$ seen as a $\mathcal{C}([0,1])$ -valued random variable, the Brownian snake driven by e is the centered Gaussian process Z^e (conditional on e) with (conditional) covariance function

$$E[Z_s^e Z_t^e | \sigma(e)] = m_e(s, t).$$

By the identification of $[0, 1]$ with $\mathbb{S}^1$ we can identify:

- The Brownian excursion $e = \{e_a\}_{a \in \mathbb{S}^1}$ can be viewed as indexed by $\mathbb{S}^1$ and a $\mathcal{C}(\mathbb{S}^1)$ -valued random variable. Moreover, we have the following quantity:

$$m_e(a, b) = \max \left\{ \min_{c \in [a, b]} e_c, \min_{c \in [b, a]} e_c \right\}, \quad a, b \in \mathbb{S}^1.$$

- Similarly, the Brownian snake Z^e driven by e can be viewed as the centered Gaussian process with the covariance function

$$E[Z_a^e Z_b^e | \sigma(e)] = m_e(a, b).$$

It is a classical result that the Brownian excursion e (see e.g. [5]) has almost surely distinct local minima, and so e satisfies the assumption (H_e) . Now I will state a technical lemma, which will help to construct the *Brownian map*. The proof can be found in [11] and [14].

Lemma 3.29 *The Brownian snake Z^e driven by e (seen as a $\mathcal{C}(\mathbb{S}^1)$ -valued random variable) satisfies the assumption (H_{Z^e}) almost surely.*

This lemma is all we need to construct the *Brownian map*.

Construction 3.30 Let $e = \{e_a\}_{a \in \mathbb{S}^1}$ be a Brownian excursion and $Z^e = \{Z_a^e\}_{a \in \mathbb{S}^1}$. Set now

$$\underline{Z} = \min_{a \in \mathbb{S}^1} Z_a^e.$$

By Lemma 3.29 there is almost surely a unique point $a^* \in \mathbb{S}^1$ such that $Z_{a^*}^e = \underline{Z}$. For $a = e^{2i\pi s}, b = e^{2i\pi t} \in \mathbb{S}^1$ set now

$$a \oplus b := \begin{cases} e^{2i\pi(s+t)}, & \text{if } s+t \leq 1, \\ e^{2i\pi(s+t-1)}, & \text{if } s+t > 1. \end{cases}$$

Then, for any $a \in \mathbb{S}^1$ we define the following random variables:

$$\bar{e}_a := e_{a^*} + e_{a^* \oplus a} - 2 \max \left\{ \min_{c \in [a^*, a^* \oplus a]} e_c, \min_{c \in [a^* \oplus a, a^*]} e_c \right\}$$

and

$$\bar{Z}_a := Z_{a^* \oplus a}^e - Z_{a^*}^e.$$

Note that $\bar{e}_1 = 0$ and $\bar{Z}_1 = 0$. Thus, the processes $\bar{e}$ and $\bar{Z}$ are $\mathcal{C}(\mathbb{S}^1)$ -valued random variables. We observe by the same argument as for the Brownian snake Z^e (see Remark 3.11), that the value of $\bar{Z}_a$ only depends on the equivalence class of a in $\mathcal{T}_{\bar{e}}$. Therefore, we may think of $\bar{Z}$ as indexed by $\mathcal{T}_{\bar{e}}$.

Let $p_e : \mathbb{S}^1 \rightarrow \mathcal{T}_e$ and $p_{\bar{e}} : \mathbb{S}^1 \rightarrow \mathcal{T}_{\bar{e}}$ denote the canonical projections. For $\alpha, \beta \in \mathcal{T}_e$ resp. $\alpha, \beta \in \mathcal{T}_{\bar{e}}$ let $[\alpha, \beta]$ denote the image under p_e resp. $p_{\bar{e}}$ of the smallest arc $[a, b]$ in $\mathbb{S}^1$ such that $p_e(a) = \alpha, p_e(b) = \beta$ resp. $p_{\bar{e}}(a) = \alpha, p_{\bar{e}}(b) = \beta$. Set for $\alpha, \beta \in \mathcal{T}_{\bar{e}}$

$$D^\circ(\alpha, \beta) := \bar{Z}_\alpha + \bar{Z}_\beta - 2 \max \left\{ \min_{\gamma \in [\alpha, \beta]} \bar{Z}_\gamma, \min_{\gamma \in [\beta, \alpha]} \bar{Z}_\gamma \right\}$$

and

$$D^*(\alpha, \beta) := \inf \left\{ \sum_{i=1}^p D^\circ(\alpha_{i-1}, \alpha_i) \mid p \geq 1, \alpha_0 = \alpha, \alpha_1, \dots, \alpha_{p-1}, \alpha_p = \beta \in \mathcal{T}_{\bar{e}} \right\}.$$

I will now show that D^* is a pseudometric.

Proposition 3.31 D^* defines a pseudometric on $\mathcal{T}_{\bar{e}}$.

Proof By Construction 3.30 D^* is non-negative and symmetric. It remains to be shown that it satisfies the triangle inequality. Let $\alpha, \beta, \gamma \in \mathcal{T}_{\bar{e}}$. Let $p, q \geq 1$ and fix finite sequences

$$\alpha_0 = \alpha, \dots, \alpha_p = \beta \in \mathcal{T}_{\bar{e}}$$

and

$$\beta_0 = \beta, \dots, \beta_q = \gamma.$$

Then, we can compute that

$$D^*(\alpha, \gamma) \leq \sum_{i=1}^p D^\circ(\alpha_{i-1}, \alpha_i) + \sum_{i=1}^q D^\circ(\beta_{i-1}, \beta_i).$$

Because these finite sequences were chosen arbitrarily, we conclude that

$$D^*(\alpha, \gamma) \leq D^*(\alpha, \beta) + D^*(\beta, \gamma).$$

This concludes the proof. $\square$

Now I will state a result given [15].

Lemma 3.32 *The quantities D° and D^* satisfy the following equivalence for all $\alpha, \beta \in \mathcal{T}_{\bar{e}}$:*

$$D^\circ(\alpha, \beta) = 0 \iff D^*(\alpha, \beta) = 0.$$

Now we are ready to define the *Brownian map*.

Definition 3.33 *On $\mathcal{T}_{\bar{e}}$ define the following equivalence relation:*

$$\alpha \approx \beta \iff D^*(\alpha, \beta) = 0 \stackrel{\text{Lemma 3.32}}{\iff} D^\circ(\alpha, \beta) = 0.$$

The **Brownian map** is defined to be the quotient space $(\mathcal{T}_{\bar{e}}/\approx, D^*)$.

This concludes this subsection on the construction of the Brownian map. We now turn our attention to the last result presented in this thesis.

3.2.3 Homeomorphism result

This last part is devoted to the result, which states that the Brownian map in Definition 3.33 is homeomorphic to the 2-sphere $\mathbb{S}^2$. To prove this, we need the following technical lemma, which is proven in [11].

Lemma 3.34 *The assumption (H_{e, Z^e}) almost surely holds, i.e. almost surely for every $a, b, c \in \mathbb{S}^1$ we have*

$$a \sim_e b \text{ and } a \sim_{Z^e} b \implies a = b \text{ or } a = c.$$

With Lemmata 3.29 and 3.34 and Proposition 3.27 the following result is obtained.

Lemma 3.35 *The quotient space $\mathcal{T}_e/\sim_{Z^e}$ is almost surely homeomorphic to the 2-sphere $\mathbb{S}^2$.*

Now, we are in the position to state and prove the main result.

Theorem 3.36 *The Brownian map $(\mathcal{T}_{\bar{e}}/\approx, D^*)$ is almost surely homeomorphic to 2-sphere $\mathbb{S}^2$.*

Proof By Lemma 3.35 it suffices to show that $(\mathcal{T}_{\bar{e}}/\approx, D^*)$ is almost surely homeomorphic to $\mathcal{T}_e/\sim_{Z^e}$. Let $\rho : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ denote the rotation with angle $2\pi s^*$, where $a^* = e^{2i\pi s^*}$. Then by Lemma 2.9, ρ induces an isometry

$$R : (\mathcal{T}_{\bar{e}}, d_{\bar{e}}) \rightarrow (\mathcal{T}_e, d_e).$$

Also R satisfies the following identity:

$$\bar{Z}_\alpha = Z_{R(\alpha)}^e - \underline{Z}, \quad \alpha \in \mathcal{T}_{\bar{e}}.$$

By Lemma 3.32 we have that almost surely for every $\alpha, \beta \in \mathcal{T}_{\bar{e}}$:

$$\begin{aligned} \alpha \approx \beta &\iff D^\circ(\alpha, \beta) = 0 \\ &\iff \bar{Z}_\alpha = \bar{Z}_\beta = \max \left\{ \min_{\gamma \in [\alpha, \beta]} \bar{Z}_\gamma, \min_{\gamma \in [\beta, \alpha]} \bar{Z}_\gamma \right\}. \end{aligned}$$

By the above, this is equivalent to

$$Z_{R(\alpha)}^e = Z_{R(\beta)}^e = \max \left\{ \min_{\gamma \in [R(\alpha), R(\beta)]} Z_\gamma^e, \min_{\gamma \in [R(\beta), R(\alpha)]} Z_\gamma^e \right\}.$$

Hence, it follows that $R(\alpha) \sim_{Z^e} R(\beta)$. This shows that R induces a bijection

$$\tilde{R} : \mathcal{T}_{\bar{e}}/\approx \rightarrow \mathcal{T}_e/\sim_{Z^e}.$$

Left to show is that $\tilde{R}^{-1}$ is continuous, as $\mathcal{T}_{\bar{e}}/\approx$ and $\mathcal{T}_e/\sim_{Z^e}$ are compact. If we show this, it follows that $\tilde{R}$ is a homeomorphism. Because the canonical projection $\mathcal{T}_{\bar{e}} \rightarrow \mathcal{T}_{\bar{e}}/\approx$ is continuous, we can compose $\tilde{R}^{-1}$ from $\mathcal{T}_e$ to $\mathcal{T}_{\bar{e}}$ with this canonical projection, which again gives a continuous map

$$\mathcal{T}_e \rightarrow \mathcal{T}_{\bar{e}}/\approx.$$

This induces then a continuous map

$$\mathcal{T}_e/\sim_{Z^e} \rightarrow \mathcal{T}_{\bar{e}}/\approx.$$

However, this is just the mapping $\tilde{R}^{-1}$. So $\tilde{R}^{-1}$ is continuous. $\square$

This concludes the proof of the main result and concludes our discussion of the Brownian map.

3.3 Final remarks

To conclude this thesis, I will shortly highlight the main points of this thesis:

- We introduced the continuum random tree as the quotient space $\mathcal{T}_e = [0,1]/\sim_e$ coded by a Brownian excursion e . This construction was justified, as the coding map $\mathcal{T}_\bullet$ is a Borel measurable map with values in the space of real trees.

3. THE BROWNIAN MAP

- We saw the connection the continuum random tree had to discrete Galton-Watson tree, as the its distributional limit.
- We explored the concept of a Brownian motion indexed by real trees, which extended to Brownian motion indexed by the continuum random tree by identifying it as the Brownian snake driven by Brownian excursion.
- This all culminated in the construction of the Brownian map and its homeomorphism to the 2-sphere via the theory of hyperbolic geodesic laminations.

This summarizes the work done in this exposition. I should add that there is much more to explore. There is another construction of the Brownian map via the scaling limits of large planar maps. This is done extensively in [9]. Intuitively, we can think of the results presented there, as randomly drawing graphs on the 2-sphere and the considering its limit as the number of vertices goes to infinity. This was skipped in the context of this semester thesis, as it would be a quite lengthy exploration, which would be a semester project in its own right. This just comes to show, the richness of the Brownian map and the continuum random tree as models in random geometry, and how random geometric structures arise in several different ways.

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